Multiscale Causation in Self-Aware Networks

Bottom-Up Construction, Top-Down Constraint, and Lateral Coordination

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Final preprint, Version 1.0 — 13 August 2026

Abstract

A nervous system can be described as receptors and cells, circuits and cortical areas, a behaving organism, or an organism coupled to an environment. Each description can be useful, yet the word *cause* is often applied across these levels without specifying what was changed. A local event may help construct a larger state; a task rule or boundary condition may constrain which local transitions remain available; peers may cooperate, compete, inhibit, synchronize, or exchange messages; and action may return consequences that alter all three processes. Treating these as one undifferentiated hierarchy hides the mechanism and makes causal claims difficult to test.

This paper develops an intervention-based framework for Self-Aware Networks (SAN) that separates **bottom-up construction**, **top-down constraint**, and **lateral coordination** while retaining their interaction through returned consequence. The model defines local state, peer topology, physically implemented context, macrostate, action, and environmental return. It then requires mapped interventions, micro–macro consistency, declared realization fibers, leakage controls, alternative coarse-grainings, and matched rival models. The SAN-specific proposal is that receiver-relative local differences can construct larger representational states; lateral phase, message, and inhibitory relations can change routing and effective drive; physically implemented context can alter receiver readiness and available transitions; and action can return evidence that updates the system.

Four preregistered synthetic studies test the measurement logic rather than assuming the framework is true. They recover implanted positive, null, and reversed effects; expose shared-input bias, leakage failure, operation-order sensitivity, weakly powered interactions, map misspecification, and oracle-to-learned-map loss; and compare the joined representation with capacity-matched flat, equivariant, relational, and trained nonlinear alternatives. The evidence is deliberately mixed. Structural maps can improve transfer efficiency under remapping, but a generic equivariant model ties in one study and a larger flat model catches up in another. Exact contrast tests support four role contrasts while three remain unresolved. Learned relations degrade under missing probes and noise. Twelve finite invariants compile in Lean, but those proofs do not establish empirical premises.

A fixed-source audit then asks whether public data already permit the complete external test. Seven initial candidates supply serious architectural, physical-intervention, lateral-intervention, or implementation evidence, but none exposes all twelve required fields for the crossed four-role endpoint. DynaTeamTHOR is the closest architectural match; its fixed archive contains 720 scenario inputs but not the raw per-episode joint-role outcomes needed for reanalysis. An independent freshness pass adds four stronger collaboration datasets or protocols and reaches the same bounded result. No missing outcome is generated or imputed. The framework therefore ends with a fully specified physical experiment, explicit failure conditions, and a sealed 8,192-block final test—not a claim of biological validation, consciousness, or universal predictive superiority.

Keywords: causal abstraction; multiscale causation; Self-Aware Networks; Phase Wave Differential; thalamocortical loops; lateral coordination; collective dynamics; intervention; coarse-graining; recurrent agency; external-source eligibility

1. The problem appears before the vocabulary

Imagine holding a light cup whose weight suddenly changes. Receptors in the hand and arm register new conditions. Local circuits transform those signals. Larger sensorimotor networks revise the estimated state of the cup and body. A selected motor route changes muscle activation. The movement changes the cup, the hand, and the next sensory input. At the same time, attention and task context change which details matter, while neighboring neural populations facilitate some routes and inhibit others.

Several causal statements can be made about this episode, but they do not all mean the same thing. A receptor event contributes to a larger pattern. A rule such as “do not spill” changes which actions remain acceptable. Neighboring populations compete and cooperate during route selection. The chosen movement changes the world, and that consequence returns to the network. A claim that “the system caused the action” compresses these distinct operations into one phrase.

The same ambiguity appears in a flock. Each bird responds to neighbors and environmental cues, yet the group forms a coherent direction that changes what any one bird encounters. It appears again in an artificial agent system. Local workers produce candidate outputs, peer messages change one another’s state, a shared project context changes available operations, and execution changes the environment that all agents subsequently observe.

The problem is therefore not whether higher and lower descriptions are physical. Every realized influence must occur through a physical or computational pathway. The scientific problem is to determine:

1. which variables exist at each declared scale;
2. which interventions are physically realizable;
3. whether a higher-level variable preserves intervention effects rather than merely summarizing observations;
4. whether peer relations add a causal effect beyond common input;
5. whether the conclusion survives alternative coarse-grainings; and
6. whether causal roles remain stable as the system learns or develops.

Only after these questions are answered does it become useful to name three causal roles.

2. Established scientific foundation

2.1 Correlation becomes causation through a declared intervention

Pearl's structural causal framework distinguishes observing a variable from setting it through an intervention [1]. Halpern further distinguishes population-level intervention effects from claims about an actual cause in one event [2]. This matters for multiscale systems because a large-scale pattern can correlate with local activity without independently supporting a meaningful intervention.

A structural causal model states how endogenous variables are generated from parents and exogenous conditions. Its value is not the diagram alone. The model earns a causal interpretation by predicting what changes under allowed interventions. Hidden common causes, feedback, selection, measurement error, and an intervention that changes several targets at once can each make an apparent causal direction unreliable [1,2,26,27].

For this paper, "top," "bottom," and "lateral" are not spatial adjectives. They are intervention roles relative to an explicitly declared partition. A cortical area can be "higher" for one representational hierarchy and a peer in another network. A thalamic nucleus can relay one signal, modulate another pathway, and participate recurrently in a third operation. The role must be indexed to variables, time, task, and intervention.

2.2 A macrostate must preserve intervention meaning

Rubenstein and colleagues formalized exact transformations between low- and high-level structural equation models [3]. The central requirement is consistency: a low-level intervention, mapped to the high-level model, should lead to the same predicted macro result as the corresponding high-level intervention. Beckers and Halpern refined this approach by distinguishing exact, uniform, strong, and allowed-intervention abstractions [4]. Beckers, Eberhardt, and Halpern then extended the framework to approximate abstractions, which are more realistic when a coarse model is useful but imperfect [5].

These results block a common shortcut. Averaging a population, naming the average, and drawing an arrow from it back to its constituents does not establish top-down causation. The higher-level variable must correspond to an implementable family of lower-level changes, and the intervention mapping must be declared. If two microstates map to one macrostate but respond differently to the same supposed macro intervention, the abstraction is incomplete for that intervention.

Hoel, Albantakis, and Tononi proposed effective information as one way to compare causal informativeness across coarse-grainings [6,37]. Under some model structures, a macro description can produce a larger value than the micro description because aggregation reduces degeneracy or indeterminism. This is a candidate scale-selection result, not a universal license for macro primacy. The value depends on system definition, intervention distribution, transition model, and coarse-graining. This paper therefore treats causal emergence as one diagnostic to compare with predictive accuracy, control value, commutation error, and robustness.

Two further questions must be answered before a macro intervention is treated as identified. First, one macro setting can correspond to many low-level realizations. Eberhardt and Lee show why an average over those interventions can conceal important heterogeneity and why the result depends on how the intervention family is weighted [38]. Second, abstraction and removal of nuisance variables need not yield the same answer in either order [38]. A macro effect must therefore report both its low-level intervention fiber and the distribution used to realize that fiber, rather than silently replacing an experimentally available intervention with a uniform average.

Other causal-emergence programs supply complementary tests. Flack treats coarse-graining as an endogenous physical process by which interacting components infer and stabilize aggregate regularities [39]. Rosas and colleagues use information decomposition to distinguish causal decoupling, downward influence, and synergistic emergence [40]. Noble argues that biological explanation should not presume one permanently privileged causal scale [41]. These accounts are not interchangeable, yet together they sharpen the same experimental requirement: specify the variables, physical realization, information retained or discarded, intervention family, and observable consequence before deciding which scale is useful.

2.3 Neural circuits already exhibit separable directional roles

Neuroscience supplies concrete cases in which a larger task context, a specific pathway, and local circuit state cannot be compressed into one generic feedback label. Zanto and colleagues perturbed human prefrontal function using repetitive transcranial magnetic stimulation. The perturbation reduced top-down modulation of posterior cortical activity during encoding and predicted worse working-memory performance [7]. The experiment did not manipulate an abstract “macro mind”; it altered a physical cortical target within a task-defined network.

Schmitt and colleagues found that mediodorsal thalamic input sustained rule representations by amplifying local prefrontal connectivity rather than simply relaying the categorical rule [8]. Broadly increasing prefrontal excitability reduced rule specificity, while increasing mediodorsal excitability improved it. That result is especially useful here: the amount of activity and the organization of activity can be dissociated.

Bolkan and colleagues used pathway-specific inhibition to separate directions in reciprocal mediodorsal-thalamic and medial-prefrontal circuitry [9]. Mediodorsal-to-prefrontal signaling supported working-memory maintenance, whereas prefrontal-to-mediodorsal signaling supported later choice. Rikhye and colleagues further showed that thalamic populations tracked contextual regularity and helped augment context-relevant prefrontal representations while suppressing irrelevant ones [10]. Pulvinar and cortical communication-subspace findings provide additional evidence that routing and communication depend on task and population geometry [11,12].

These experiments establish a scientific foundation for direction-sensitive, context-sensitive causal decomposition. They also show why “top-down” must be tied to anatomy and manipulation. The high-level task rule is realized by distributed activity and pathways. It constrains local transitions through those implementations rather than acting as an extra physical force.

2.4 Collective order can be constructed by local rules

Reynolds demonstrated that simple local rules could produce convincing artificial flocking [14]. Vicsek and colleagues introduced a self-propelled-particle model in which local neighbor alignment, motion, and noise produced a transition from disordered motion to macroscopic order [13]. Couzin and colleagues modeled how individual interaction rules can produce collective memory and sorting [15]. Measurements of natural bird flocks have revealed structured correlations and interaction statistics, although observation alone does not identify every causal edge [16,17].

Oscillator research provides a complementary case. Winfree and Kuramoto showed how local phase coupling can generate collective timing [18,19]. Strogatz clarified how order parameters summarize synchronization transitions [20]. O’Keeffe, Hong, and Strogatz coupled phase and spatial motion in “swarmalator” models, yielding several collective regimes [21]. Peleg and colleagues reconstructed and analyzed natural firefly synchrony, including self-organization and mixed synchronized–unsynchronized chimera states [22–24].

These systems matter because they make two distinctions visible. First, local interaction can construct a macrostate without a central controller. Second, the macrostate can change the conditions encountered by individuals, but that return must be physically specified—for example through local density, visual exposure, boundary geometry, resource distribution, or neighbor availability. Group order, semantic representation, cognition, and consciousness remain different claim classes.

2.5 Identification, interference, and informative null results

An instruction to perturb a system is not yet the perturbation that the system received. A light pulse can miss a cell class, a stimulation pulse can spread beyond its intended pathway, a boundary change can alter both visibility and density, and a message-graph intervention can change load as well as topology. Prospective Analysis Plan therefore separates **assignment**, **realized exposure**, **intervention fidelity**, and **outcome**. An intention-to-treat effect answers what the randomized assignment changed. A realized-exposure effect answers a different question and requires additional assumptions when assignment and exposure diverge.

For a treatment-effect estimate to identify the stated causal contrast, the analysis must declare at least: consistency between the named intervention and the corresponding potential outcome; exchangeability supplied by randomization or an adequate adjustment strategy; positive probability for every intervention level within the analyzed support; a time order that does not condition on descendants of treatment improperly; measured loss, censoring, and intervention failure; and an interference model. These requirements are standard causal discipline rather than SAN-specific assumptions [1,26,27,42,48,49].

The usual no-interference simplification is unsuitable for a theory whose subject includes peer influence. Neural populations, animals in a group, and communicating agents can change one another's outcomes. Hudgens and Halloran show how direct, indirect, total, and overall effects can be defined under group-structured interference [42]. Aronow and Samii formalize exposure mappings for more general interference structures [49]. Paper 9 therefore does not pretend that units are isolated. It preregisters which neighbors can matter, defines an exposure mapping, randomizes at a level compatible with that mapping, and treats unmodeled spillover as a failure of identification rather than as unexplained noise.

Mediation requires similar restraint. A randomized context intervention does not automatically identify how much of its effect passed through peer organization or receiver state. Natural direct and indirect effects require strong cross-world or sequential-ignorability assumptions [43]. Interventional direct and indirect effects replace an individual's counterfactual mediator with a draw from a declared mediator distribution and can be identified under weaker, still explicit conditions [44]. Prospective Analysis Plan makes that interventional decomposition the default. A path-specific interpretation is promoted only when the mediator is itself randomized or its mediator–outcome confounding is adequately controlled and sensitivity-tested.

Finally, a nonsignificant estimate is not evidence that two conditions are equivalent. Equivalence requires a smallest effect size of interest fixed before held-out analysis and uncertainty narrow enough to exclude effects larger than that boundary [45]. Underpowered estimates can also exaggerate magnitude or even return the wrong sign [46]. The application program will therefore use prospective design calculations and simulation-based power analysis with declared data-generating mechanisms, estimands, methods, and performance measures [47]. Noise can force an unresolved result; it cannot enlarge the scientific tolerance after outcomes are known.

3. The SAN genealogy: from nested loops to an explicit causal taxonomy

The current framework did not appear all at once. The dated source ladder preserves partial ancestors rather than assigning them the later vocabulary.

3.1 2011: distributed construction and returned consequence

An internally dated May 2011 note described small distributed sensory elements and learned statistical relations combining into a continuously updated multisensory picture [28]. This is an early bottom-up construction ancestor: small patterns contribute to larger patterns, and the larger representation supports prediction. The note does not yet define causal abstraction or the later three-role taxonomy. Its 2011 date is internal; its public Git fixation occurred in 2022.

A second note, bounded to before September 21, 2011, joined statistical memory and prediction to brain activity, muscular control, sensory return, and a learning loop that learns itself [29]. This supplies the early recurrent arc:

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learned prediction -> action -> changed sensory input -> updated loop
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It does not yet separate lateral coordination from top-down constraint. It establishes that action and returned evidence were part of the architecture before the later terminology.

3.2 2012: hierarchy, inhibition, morphology, and multidirectional traffic

An internally dated September 2012 exchange argued that inhibition could alter lower-level availability, that dendritic and synaptic morphology mattered to transmission, and that a small local difference could change a larger represented state [30]. An October 2012 note described cortical traffic in vertical, horizontal, front-to-back, back-to-front, and recurrent directions, while connecting learned associations to causes, motor controls, consequences, and inhibition [30].

These sources contain multilevel and lateral-routing ancestors. They do not prove that every small cellular event influences every other cell through a direct edge. Their stronger recoverable operation is that a local event can change downstream population state and represented consequence through network paths, while hierarchical and inhibitory context changes which paths remain effective.

3.3 2017: a public neural-interface bridge and an episode-level nested-loop source

The June 2017 Neural Lace Podcast program and its public article fixed an important bridge [31]. Micah Blumberg's public article identifies the episode, names its cohosts, describes the brain as a special kind of hard drive, and proposes reading the current neural state before introducing only the difference needed for a target brain-computer-interface state. The local episode transcript separately contains a passage about small and large feedback loops, thalamus-to-cortex-to-thalamus recurrence, and a brain-scale fractal of feedback loops.

The public article dates the program and establishes Micah's authored neural-interface formulation. The transcript supplies the spoken nested-loop wording but was fixed locally later and does not diarize the relevant passage well enough to assign that exact wording to one speaker. The paper therefore cites it at episode level. Keeping authorship, custody, and locator layers separate allows both components to be used without overstating the provenance of either sentence.

3.4 2021: a recorded flow-of-information bridge

The March 25, 2021 *Flow of Information in the Brain* recording supplies a further source stage [36]. Its dialogue describes temporal and spatial properties of broccoli as an example of how differently organized brains can recognize a shared object, follows sensory traffic through thalamic and cortical pathways, and discusses vertical, horizontal, feedforward, feedback, and recurrent paths as nested loops. It also distinguishes local recurrent circuits from larger oscillatory and whole-brain descriptions.

The recovered transcript contains an original Google Recorder share URL and states the recording date, but the service metadata was not independently read back for Source-and-Schema Audit. It is therefore treated as an owner-held recorded-source bridge with an exact local hash, not as independent public fixation of every utterance. The dialogue is also incompletely diarized. This 2021 source strengthens the traffic-and-scale genealogy; it does not receive the top-down/bottom-up/lateral terminology first fixed in the reviewed January 2024 source.

3.5 2022: receiver arrays, population scaling, PWD, and coordination

The 2022 SAN Git record names Neural Array Projection Oscillation Tomography (NAPOT) and develops receiver arrays that combine multiple inputs, transform them according to learned sensitivity, and project updated patterns onward [32]. One note describes a group representation that can vote, correct, and remain robust to the loss or substitution of a single element. It uses fireflies as an analogy for synchronized group sensing while leaving the biological details open.

A July note asks how cellular events can be tracked by larger oscillating assemblies and how frequency, timing, inhibition, and route bifurcation can change distributed choice [32]. An August note traces learned object properties from receptors through cells, arrays, thalamic and cortical loops, and back through recurrent pathways. It names phase-wave differentials as receiver-relevant changes against an ongoing context and connects temporal alignment to cross-source association [32].

This stage supplies a mature local-to-population operator, but it still lacks the explicit three-direction causal taxonomy and the intervention discipline developed below.

3.6 2023–2024: scaling and the named three-role program

The 2023 whitepaper states that memory and selective reception scale from synaptic and dendritic structure through arrays and wider brain activity [33]. It describes a neuron as both a selective receiver and a projector whose output changes the group, while the group's ongoing state changes how the event is registered.

The January 2024 SAOv9 source then names "causation from all angles" and explicitly lists top-down, bottom-up, and lateral causation, their integration in Self-Aware Networks, their effects on artificial behavior, ethical consequences, evolution, and future research [34]. This is the first explicit three-role program in the frozen source set. It is an outline, not a completed causal model.

3.7 2026: matched interventions and cross-domain tests

The governed 2026 recovery decomposes the outline into testable atoms [35]. It requires physical interventions, matched activity, causal abstraction, mediation, alternative coarse-grainings, longitudinal checkpoints, and failure conditions. It also supplies the joined SAN proposal:

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bottom-up PWD construction
-> lateral phase, message, and inhibitory coordination
-> top-down contextual constraint
-> action
-> returned consequence
-> altered future roles
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The current paper formalizes that chain. The 2026 wording is a present synthesis and does not move the earlier dates backward.

4. Earning the three causal terms

4.1 Bottom-up construction

Suppose local elements have states and interactions. If changing a declared subset changes a larger-scale state under a controlled intervention, those local elements participate in constructing that macrostate. Construction can involve summation, nonlinear transformation, competition, thresholding, routing, or temporal integration. It need not mean that every component contributes equally.

Bottom-up construction is therefore the interventionally measured contribution of lower-level states and relations to a declared higher-level variable. The macrostate may be an order parameter, a decoded representation, a task rule state, an action policy, or a system-level output. The mapping from local states to the macrostate must be explicit.

4.2 Top-down constraint

A task rule can change which sensory evidence is relevant. A boundary can change which routes an agent can traverse. A shared state can change which operations an artificial worker is authorized to perform. In each case, the larger context is implemented by physical or computational variables.

Top-down constraint is the effect of an implemented context, rule, boundary, macrostate, or control pathway on lower-level transition probabilities or available routes. The term is earned only when the implementation can be intervened on and when its effect is not exhausted by an uncontrolled change in common input or total activity.

Constraint is the right word because the macro description usually selects, weights, opens, closes, amplifies, or suppresses possibilities already realized by lower-level dynamics. It does not suspend those dynamics.

4.3 Lateral coordination

Two peers can exchange messages, align timing, compete for control, inhibit one another, or cooperate on a shared representation. Their apparent coordination may also arise because both receive the same hidden input.

Lateral coordination is the causal effect of interactions among elements at a comparable declared scale after common input and shared context are controlled. Direction, delay, sign, and topology are part of the definition. A correlated pair is not enough.

4.4 Recurrent return

The three roles are not a one-way ladder. Action changes an environment or body; the changed environment alters later local inputs; learning changes peer topology and receiver tuning; and repeated context changes can reorganize the macrostate itself. **Recurrent return** names this consequence-to-next-state path. It prevents a static hierarchy from masquerading as an agent.

5. A multiscale causal model

5.1 State variables

Let $X_t = (x_{1t}, \ldots, x_{nt})$ denote local states, A_t the directed peer-coupling graph, B_t the physically implemented context or boundary state, E_t the environment or body state, M_t a declared macrostate, and Y_t the system output or action. Exogenous disturbances are denoted U_t.

The local dynamics are:

$$X_{t+1} = F_X(X_t, A_t, B_t, E_t, U_t^X). \tag{1}$$

Peer topology can itself learn or change:

$$A_{t+1} = F_A(A_t, X_t, B_t, E_t, U_t^A). \tag{2}$$

The macrostate is an explicit coarse-graining rather than a second substance:

$$M_t = \tau(X_t, A_t, B_t). \tag{3}$$

Action depends on the state used by the controller:

$$Y_t = F_Y(X_t, A_t, B_t, M_t, E_t, U_t^Y). \tag{4}$$

Action changes the environment and therefore future input:

$$E_{t+1} = F_E(E_t, Y_t, U_t^E). \tag{5}$$

Equations (1)–(5) define the minimum closed loop. Removing Equation (5) produces a recurrent processor with no modeled consequence return. Treating M_t as causally active without specifying how B_t, X_t, or A_t implement its influence produces an ungrounded macro arrow.

5.2 Receiver-conditioned difference

SAN begins from a practical fact: the effect of an arriving event depends on the receiver's current condition. Let $s_{i,t}$ be the arriving signal, $h_{i,t}$ the receiver's recent state and learned sensitivity, and $b_{i,t}$ the context available at that receiver. The transformed event is:

$$q_{i,t} = T_i(s_{i,t}, h_{i,t}, b_{i,t}). \tag{6}$$

The receiver-relative departure from its expected ongoing relation is:

$$\delta_{i,t} = q_{i,t} - \mathbb{E}[q_{i,t} \mid h_{i,t}, b_{i,t}, E_t]. \tag{7}$$

The scalar form in Equation (7) is a convenient projection. In an empirical implementation, $\delta_{i,t}$ may be a vector containing timing, circular phase, waveform duration, amplitude, spatial relation, excitation/inhibition sign, and downstream consequence. SAN calls the operational receiver-relative departure a **Phase Wave Differential (PWD)** after the receiver, expectation, and consequence are specified.

A larger PWD-related state is constructed by a declared map:

$$M_t^{\mathrm{PWD}} = \tau_{\mathrm{PWD}}\big(\{\delta_{i,t}\}_{i=1}^{n}, A_t, B_t\big). \tag{8}$$

Equation (8) is a testable composition claim. It does not follow merely because local differences and a population signal are both observed.

5.3 Lateral phase and message coordination

For oscillatory components, a phase-sensitive peer model can be written:

$$\dot{\phi}_i = \omega_i + \sum_j A_{ij,t} K_{ij,t} \sin(\phi_j - \phi_i - \alpha_{ij}) + u_i(B_t, E_t). \tag{9}$$

Here $A_{ij,t}$ states whether a route exists, $K_{ij,t}$ its effective strength, α_{ij} a preferred phase relation, and u_i a contextual or environmental drive. This extends the classical comparison only enough to represent directed, signed, and context-sensitive coupling.

The effective drive received by a downstream component can be estimated as:

$$D_{i,t} = \sum_j A_{ij,t} w_{ij,t}\, g(\delta_{j,t})\ \cos(\phi_j - \phi_i - \alpha_{ij}), \tag{10}$$

where g maps the receiver-relevant difference into the measured input dimension. Equation (10) makes two experimentally separable claims: organization can change effective drive while total activity is matched, and total activity can change while organization is held approximately fixed.

5.4 Role-specific intervention effects

For a prespecified set S of local units, define the bottom-up construction effect on the next macrostate:

$$\Delta_{\mathrm{BU}}(S) = \mathbb{E}[M_{t+1} \mid do(X_{S,t} = x^{(1)})] - \mathbb{E}[M_{t+1} \mid do(X_{S,t} = x^{(0)})]. \tag{11}$$

For a physically implemented context B_t, define a top-down constraint effect on local transitions under matched starting state and environment:

$$\Delta_{\mathrm{TD}} = \mathbb{E}[X_{t+1} \mid do(B_t = b^{(1)}), \mathcal{M}_t] - \mathbb{E}[X_{t+1} \mid do(B_t = b^{(0)}), \mathcal{M}_t], \tag{12}$$

where $\mathcal{M}_t$ denotes the preregistered matching set for local state, environment, task difficulty, and total activity. The context intervention must be implemented through named lower-level nodes or boundaries.

For peer topology or message state A_t, define the lateral effect on a common outcome O:

$$\Delta_{\mathrm{LAT}} = \mathbb{E}[O \mid do(A_t = a^{(1)}), \mathcal{L}_t] - \mathbb{E}[O \mid do(A_t = a^{(0)}), \mathcal{L}_t], \tag{13}$$

where $\mathcal{L}_t$ matches local input, context, environment, and aggregate activity. The three results form a time-indexed role vector:

$$\mathbf{R}_t = (\Delta_{\mathrm{BU}}, \Delta_{\mathrm{TD}}, \Delta_{\mathrm{LAT}}, \Delta_{\mathrm{RET}}), \tag{14}$$

with $\Delta_{\mathbf{RET}}$ measuring the effect of changed action consequences on later state. A system can have all four nonzero entries. The taxonomy separates roles; it does not force one level to win.

5.5 Micro–macro consistency

Let $\mathcal{M}_L$ be a low-level causal model, $\mathcal{M}_H$ a high-level model, τ the state map, and ω the intervention map. Exact causal consistency requires:

$$\tau_{\#} P_L^{do(i)} = P_H^{do(\omega(i))} \quad \text{for every allowed } i, \tag{15}$$

where $\tau_{\#}$ is the push-forward distribution. For empirical systems, define abstraction error:

$$\varepsilon_\tau = \sup_{i \in \mathcal{I}_L^{\text{allowed}}} D\Big(\tau_{\#} P_L^{do(i)}, P_H^{do(\omega(i))}\Big), \tag{16}$$

with a preregistered distance D. A macro model is useful for the intervention family only if ε_τ is acceptably small relative to baselines and uncertainty.

5.6 Causal informativeness, mediation, and leakage

One candidate scale diagnostic is effective information:

$$EI(M) = I(do(M_t); M_{t+1}), \tag{17}$$

under a declared intervention distribution. A normalized macro–micro difference can be reported as:

$$CE_\tau = \frac{EI(M)}{\dim(M)} - \frac{EI(X)}{\dim(X)}. \tag{18}$$

Positive CE_τ is evidence that a particular coarse description is efficient under the chosen metric and intervention distribution. It is interpreted alongside Equation (16), not as a free-standing metaphysical conclusion.

To determine whether a context effect is mediated by peer organization or receiver state, decompose the total effect:

$$TE_{B\to O} = NDE_{B\to O} + NIE_{B\to(A,X)\to O}, \tag{19}$$

under the assumptions required by the chosen mediation estimator. Because interventions may spill across roles, report a leakage score:

$$\Lambda_k = \frac{\|\Delta Z_{\neg k}\|}{\|\Delta Z_k\| + \epsilon_0}, \tag{20}$$

where Z_k is the intended target family and $Z_{\neg k}$ all monitored non-target families. High leakage prevents a clean role assignment.

5.7 Robustness across scales and learning

Let $\mathcal{T} = \{\tau_1, \ldots, \tau_K\}$ be plausible coarse-grainings chosen before outcome inspection. Define role robustness:

$$\rho_k = 1 - \frac{\mathrm{Var}_{\tau \in \mathcal{T}}(\widehat{\Delta}_{k,\tau})}{\mathrm{Var}_{\tau \in \mathcal{T}}(\widehat{\Delta}_{k,\tau}) + \sigma_k^2}, \tag{21}$$

where σ_k^2 is a declared reference variance. This is a diagnostic, not a universal statistic; the preregistration must specify the estimand and uncertainty model.

Finally, learning can move causal influence among roles. Between checkpoints t_a and t_b, define:

$$\Gamma(t_a, t_b) = \|\mathbf{R}_{t_b} - \mathbf{R}_{t_a}\|_{\Sigma^{-1}}, \tag{22}$$

where the covariance-weighted norm distinguishes genuine role migration from estimation noise. A static causal label is inadequate when Γ is large.

5.8 Macro-intervention fibers and operation order

A macro intervention label can hide several low-level ways of realizing the same declared setting. Let j be a high-level intervention and define its allowed low-level intervention fiber:

$$\mathcal{F}(j) = \{i \in \mathcal{I}_L^{\mathrm{allowed}} : \omega(i) = j\}. \tag{23}$$

The high-level interventional distribution is not identified until the experiment declares how realizations inside that fiber are selected. With a preregistered or empirically measured realization distribution π_j, the corresponding mixture is:

$$P_{H,\pi_j}^{do(j)} := \sum_{i \in \mathcal{F}(j)} \pi_j(i)\, \tau_\# P_L^{do(i)}. \tag{24}$$

Uniform weighting is one possible design, not a default fact about the system. To expose a macro label that combines low-level interventions with different consequences, define fiber heterogeneity:

$$H_\tau(j) = \sup_{i,i' \in \mathcal{F}(j)} D\Big(\tau_\# P_L^{do(i)}, \tau_\# P_L^{do(i')}\Big). \tag{25}$$

When (H_{\tau}(j)) is large relative to the declared tolerance, the macro effect must be reported as realization-dependent or the macro variable must be refined. Source-and-Schema Audit also tests whether eliminating nuisance variables before abstraction gives the same result as abstracting before eliminating their mapped macro representation. For a low-level nuisance family C and mapped macro family $\bar{C}$, define the operation-order error:

$$\chi_{\tau,C}(i) = D\Big(\tau_\#[\mathrm{Marg}_C\, P_L^{do(i)}], \mathrm{Marg}_{\bar{C}}[\tau_\# P_L^{do(i)}]\Big). \tag{26}$$

Equation (26) is zero only when the two declared operations commute under the chosen map and intervention. Together, Equations (23)-(26) prevent a coarse label, an arbitrary average, or a favorable order of analysis from standing in for one experimentally identified macro cause [38].

The two terms in Equation (26) must be expressed on the same retained macro outcome space. The first push-forward therefore uses the restriction of τ to the retained low-level variables, and the second marginalizes the corresponding mapped macro family $\bar{C}$. If no such compatible restriction can be specified, the operation-order comparison is undefined rather than zero.

5.9 Assignment, exposure, and role-effect identification

Let $W_{k,t}$ denote randomized assignment to an intervention intended to test role $k \in \{\mathrm{BU}, \mathrm{TD}, \mathrm{LAT}, \mathrm{RET}\}$, let $Z_{k,t}$ denote its measured realization, and let C_t contain only prespecified pre-intervention covariates. Because peers can be affected by one another, unit i's exposure includes its own assignment and the relevant pattern among neighbors:

$$G_{i,k,t} = g_{i,k}(W_{k,t}, W_{-i,k,t}, A_t, E_t^{\mathrm{pre}}). \tag{27}$$

The map $g_{i,k}$ is part of the estimand. In a clustered neural experiment it may summarize the number, identity, distance, and timing of perturbed presynaptic or peer populations. In a flock it may represent visible perturbed neighbors. In an agent system it can be the signed, delayed messages actually received. If two exposure patterns that matter to the outcome are collapsed into one value, the result is identified only for that collapsed contrast.

Under randomized assignment, consistency, the declared exposure mapping, measured attrition, and analysis at the randomization unit, the role-specific intention-to-treat contrast is:

$$\Delta_k^{\mathrm{ITT}} = \mathbb{E}[O_k \mid W_k = 1] - \mathbb{E}[O_k \mid W_k = 0]. \tag{28}$$

This contrast remains valid when realization is imperfect, but it estimates assignment rather than a pure realized-dose effect. Fidelity is recorded in the units of the targeted change. With $z_k^\star$ the intended realization, a bounded fidelity score is:

$$F_k = \max\left\{0, 1 - \frac{d_k(Z_k, z_k^\star)}{c_k}\right\}, \tag{29}$$

where d_k and the scientifically meaningful scale $c_k > 0$ are frozen before held-out analysis. A per-protocol or dose-response claim requires exchangeability for realized Z_k, not merely randomization of W_k. An instrumental-variable interpretation additionally requires relevance, exclusion, independence, and an explicitly stated target population; Prospective Analysis Plan does not assume those conditions by default.

When observational adjustment or weighted realization mixtures are used, positivity is a first-order gate:

$$\eta_k = \inf_{c,g \in \mathcal{G}_k^{\text{target}}} p_k(g \mid c) > 0, \tag{30}$$

Here p_k is the probability mass or density relative to the declared exposure measure. Finite data replace strict positivity with a preregistered minimum support threshold and weight diagnostic. A result outside supported exposure regions is extrapolation, not an identified effect. Trimming or coarsening changes the target population and must be reported as such [48].

The four roles require different identification contracts:

Role	Randomized object	Outcome scale	Required control	Identification-breaking counterexample
Bottom-up construction	local receiver state, input, or subset assignment	prespecified later macrostate	fixed or jointly randomized τ; declared exposure mapping	the perturbation changes context or peer graph before the macro readout
Top-down constraint	physical context-maintaining pathway, broadcast nodes, or boundary actuator	later local transitions and behavior	baseline state; yoked or factorial activity manipulation; leakage monitor	the "context" variable is only a post hoc readout of the same local state
Lateral coordination	peer edge, delay, visibility, timing, or message assignment	recipient or group outcome	shared-input randomization/yoking; load and total-activity factorial	apparent peer influence is induced by an unmeasured common driver
Returned consequence	environment/body consequence under a replayed or randomized action	later receiver, graph, context, or behavior	action replay; history and pre-return state	action differs across return conditions or the return is never measured

Total activity and message load are often post-treatment variables—that is, variables measured after the intervention can change them. Prospective Analysis Plan therefore does not "control them away" through an ordinary regression and call the residual causal. Organization and amount are separated by factorial design, calibrated stimulation, or yoked replay. If amount is affected by the intervention and is scientifically part of the route, it is analyzed as a mediator.

5.10 Interventional mediation and realizable macro interventions

Let Q denote a mediator family such as peer organization or receiver state, and let G_a be a stochastic draw from the measured mediator distribution under assignment a, conditional only on allowed baseline covariates. Prospective Analysis Plan uses interventional rather than natural direct and indirect effects as the default:

$$IDE_k = \mathbb{E}[O_k(1, G_0)] - \mathbb{E}[O_k(0, G_0)], \tag{31}$$

$$IIE_k = \mathbb{E}[O_k(1, G_1)] - \mathbb{E}[O_k(1, G_0)]. \tag{32}$$

These effects retain a clear experimental interpretation while avoiding an unnecessary individual-level cross-world mediator claim [44]. Identification still requires consistency, positivity, no unmeasured assignment–outcome confounding, and adequate control of mediator–outcome confounding for the selected interventional estimand. If the context intervention changes a mediator–outcome confounder, the analysis uses a longitudinal g-formula, a separately randomized mediator, or reports the mediation path as unresolved rather than silently invoking sequential ignorability [43,44].

A macro intervention j is physically realizable only when its low-level fiber is nonempty, its intended mixture has experimental support, and its actuation and monitoring contract exists:

$$\mathcal{R}(j) = \mathbf{1}\left[\mathcal{F}(j) \neq \varnothing, \min_{i:\pi_j(i)>0} p_{\text{exp}}(i \mid j) \geq p_{\min}, \mathcal{A}_j \neq \varnothing, \mathcal{Q}_j \neq \varnothing\right]. \tag{33}$$

Here $\mathcal{A}_j$ is a named actuator protocol and $\mathcal{Q}_j$ a named measurement-and-quality-control protocol. Every protocol records target, actuator, dose, timing, exposure map, monitor, prohibited collateral changes, reversibility or washout, fidelity, and leakage. If $\mathcal{R}(j) = 0$, the macro intervention is a formal label awaiting an experimental realization, not an empirical cause.

5.11 Distance choice, tolerance calibration, and power

Equations (16), (25), and (26) are distributional comparisons, so one distance cannot be inherited mechanically across all outcomes. Prospective Analysis Plan freezes the metric by outcome type:

$$D_{\mathcal{Y}}(P,Q) = \begin{cases} D_{\text{TV}}(P,Q), & \mathcal{Y} \text{ categorical,} \\ W_1(P,Q)/s_{\mathcal{Y}}, & \mathcal{Y} \text{ ordered and metric,} \\ \text{MMD}_\kappa(P,Q)/s_\kappa, & \mathcal{Y} \text{ multivariate or functional,} \end{cases} \tag{34}$$

where total variation, scaled 1-Wasserstein distance, or a characteristic-kernel maximum mean discrepancy is chosen before held-out analysis. The scales $s_{\mathcal{Y}}$ and s_κ, kernel, bandwidth rule, and any dimensional reduction are fit on training/calibration data only. Results from different metrics are not averaged into one favorable score; the primary metric and sensitivity metrics remain separate.

For a signed role effect θ_k, let $[L_k, U_k]$ be a multiplicity-adjusted confidence interval and $\delta_k^\star > 0$ the smallest consequence-relevant effect in the predicted direction. The outcome class is:

$$\mathcal{C}(\theta_k) = \begin{cases} \text{support}, & L_k > \delta_k^\star, \\ \text{practical equivalence}, & -\delta_k^\star < L_k \le U_k < \delta_k^\star, \\ \text{adverse/reversal}, & U_k < -\delta_k^\star, \\ \text{unresolved}, & \text{otherwise}. \end{cases} \tag{35}$$

For nonnegative error quantities ε_τ, H_τ, and $\chi_{\tau,C}$, the corresponding abstraction claim passes only when its simultaneous upper confidence bound is no greater than its prespecified practical bound. It fails when the lower bound exceeds that bound and is unresolved when the interval crosses it. A point estimate below tolerance is not enough.

The calibration split estimates a null measurement-and-estimation noise floor without changing the practical bound:

$$\nu_D = q_{1-\alpha}^{\text{cal}} \left\{ D(\widehat{P}^{(r)}, \widehat{P}^{(r')}) : r, r' \text{ are exchangeable null replicates} \right\}. \tag{36}$$

If ν_D exceeds the practical tolerance, the design cannot resolve the proposed equivalence. The response is more information, a better measurement, or a narrower claim—not a wider post hoc tolerance. Calibration data can set variance, kernel, and quality-control parameters; held-out data cannot select them.

The primary family contains the four role effects plus preregistered abstraction, fiber, and order diagnostics. A conservative fallback allocates:

$$\alpha^\star = \frac{\alpha_{\text{family}}}{m_{\text{primary}}}, \tag{37}$$

while a prespecified max-statistic randomization or bootstrap procedure may recover power using the observed dependence among estimands. Secondary analyses use false-discovery-rate control and remain labeled secondary. No correction method is chosen after seeing which one produces the preferred conclusion.

For an approximately Gaussian two-arm contrast with common variance σ^2, a transparent first-pass per-arm sample size is:

$$n_{\text{arm}} \ge \frac{2\sigma^2 \left(z_{1-\alpha^\star/2} + z_{1-\beta}\right)^2}{(\delta_k^\star)^2}. \tag{38}$$

Equivalence uses the one-sided critical value and the distance from the planning value to the nearest equivalence boundary [45]. Clustered or repeated systems reduce the independent information. With average cluster size m_c and intracluster correlation ρ_c, the first-pass effective size is:

$$n_{\text{eff}} = \frac{n_{\text{raw}}}{1 + (m_c - 1)\rho_c}. \tag{39}$$

Equations (38) and (39) are planning approximations, not the final design. The application proof will use a blinded simulation-based design analysis across weak, minimum-relevant, and stronger effects; low and high interference; intervention failure; leakage; graph density; learning drift; and model misspecification. It will report power, interval coverage, bias, root-mean-square error, type-I error, equivalence accuracy, Type-S probability, Type-M exaggeration, and Monte Carlo error [46,47]. The number of Monte Carlo replicates is increased until the Monte Carlo standard error for each primary error-rate estimate is below 0.005.

5.12 Explicit causal countermodels and capacity-matched alternatives

The strongest test of the joined mechanism is not an underpowered ablation. It is a set of alternatives allowed to explain the same observations with the same information and comparable capacity.

1. **Flat-micro sufficiency:** the complete low-level state predicts and controls every outcome; the macrostate adds no held-out value.
2. **Readout-only macrostate:** M_t summarizes X_t, A_t, B_t but has no separately realizable intervention.
3. **Shared-input countermodel:** an unobserved or omitted driver changes peers and outcomes, producing an apparent lateral effect.
4. **Activity-dose countermodel:** total activity or message load explains the effect attributed to organization.
5. **Latent-policy countermodel:** an unmeasured controller state produces both the context label and the selected action.
6. **Intervention-leakage countermodel:** collateral changes, not the intended target, produce the observed outcome.
7. **Realization-mixture countermodel:** a reported macro effect is created by favorable weighting over a heterogeneous fiber.
8. **Analysis-order countermodel:** the conclusion appears only after one favorable sequence of marginalization and abstraction.
9. **Synchrony-only countermodel:** phase concentration or band power explains the result without receiver-relative PWD features.

10. **Generic predictive-state countermodel:** a recurrent state-space model captures history and feedback without the three-role decomposition.
11. **Graph/message-passing countermodel:** a standard directed graph model explains routing and peer effects without SAN-specific variables.
12. **No-return countermodel:** recurrence alone explains performance; changing environmental return adds no effect.

The flat, hierarchical state-space, graph/message-passing, synchrony, recurrent, and SAN models receive the same train/calibration/held-out splits, observations, action opportunities, optimization steps, tuning budget, seed set, and logging. Trainable parameters are matched within one percent where architectures permit; otherwise performance is plotted against parameter count and measured compute over a prespecified capacity curve. A simpler model that matches performance wins the parsimony comparison. A larger comparator is also reported to test whether any SAN advantage disappears with ordinary capacity. Negative, null, reversed, or capacity-sensitive outcomes remain in the result ledger.

6. The joined SAN mechanism

The equations earn a compact mechanistic account.

First, a receiver does not process an input in isolation. Its morphology, learned connections, recent state, inhibition, and context determine how the input changes it. The receiver-relative departure in Equation (7) captures the measurable difference between what arrived and what that receiver expected or maintained.

Second, local departures are transformed and combined. Some are suppressed; some align in timing; some compete; some recruit a larger assembly. This is bottom-up construction, quantified by Equation (11), only when changing local elements changes the macrostate.

Third, peers alter one another. Phase relations, inhibitory edges, message delays, and topology change effective drive and routing. This is lateral coordination, measured by Equation (13), only after shared input is controlled.

Fourth, a task rule, bodily state, learned expectation, thalamic gate, boundary condition, or shared context changes receiver readiness and the set of available transitions. This is top-down constraint, measured by Equation (12), because the context is implemented and manipulated through named pathways.

Fifth, the system acts. The action changes the body or environment, and that consequence returns as new evidence through Equation (5). Learning then changes receiver sensitivity, peer topology, context use, and the coarse state itself.

The complete loop is therefore:

```
receiver-conditioned local difference
-> bottom-up construction
-> lateral routing and coordination
-> context-dependent constraint
-> selected action
-> returned consequence
-> changed receiver, topology, and context
```

The ordering is functional rather than a rigid clock sequence. The roles can overlap in time and recur at nested scales. Their scientific value comes from separate interventions and common outcome measures.

7. Three domains, one test grammar

7.1 Thalamocortical decision and attention loops

A neural experiment can combine Neuropixels or high-density electrophysiology with pathway-specific perturbation in a task that switches sensory rules. The local state X_t consists of unit and population features in thalamus and cortex. The lateral graph A_t contains effective within-area and peer-population relations. The implemented context B_t is the experimentally cued rule and the thalamocortical populations that maintain, amplify, or suppress it. The macrostate M_t is a preregistered rule or decision representation, and Y_t is choice.

The decisive design is a matched three-arm perturbation:

1. perturb a local sensory or receiver population while preserving rule context;
2. perturb the context-maintaining pathway while matching aggregate cortical activity;
3. perturb lateral timing or inhibitory coupling while preserving local input and rule cue.

The analysis asks whether each arm produces a distinct effect vector, whether mediation routes are separable, and whether PWD-aware receiver features improve held-out prediction beyond firing rate, band power, ordinary phase coding, and generic prediction-error baselines.

7.2 Firefly, flock, and swarm systems

Collective systems offer visible state variables and tractable simulations. In a flock model, local heading and sensory state form X_t, neighbor relations form A_t, boundaries or global environmental gradients form B_t, group direction or polarization forms M_t, and movement changes later neighbor exposure through E_{t+1}.

In fireflies, the local oscillator phase and flash state form X_t; visibility and coupling form A_t; terrain, vegetation, illumination, or experimental enclosure form B_t; and group periodicity or chimera structure forms M_t. Perturbations can alter individual timing, visibility links, or boundary conditions. The analysis must preserve species, mating context, and measured behavior. It does not assign group cognition merely because coordination is striking.

The value of these systems is comparative. A local-alignment model can test bottom-up construction. Boundary changes can test physically implemented constraint. Occlusion or neighbor manipulation can test lateral coupling. The same abstraction and leakage diagnostics used in the neural and artificial domains can then be applied without asserting biological identity.

7.3 Artificial agents and shared-context systems

Artificial systems make interventions reproducible. A reference application will use local agents with noisy observations, a directed peer graph, a shared-context register implemented by broadcast nodes, a rule-switching environment, actions, and returned consequences. Every state transition and message will be logged.

The system will compare:

- local observation clamps or substitutions;
- physical context-register changes with total activation matched;
- peer-edge delay, silencing, rewiring, or sign changes;
- environment replay with matched actions;
- multiple coarse-graining maps;
- five longitudinal learning checkpoints.

The application proof succeeds only if role-specific effects are recovered on held-out runs, the mapped micro and macro interventions approximately commute, and the joined model improves prediction or control beyond capacity-matched alternatives. Assignment, realized exposure, fidelity, interference, leakage, and return are logged separately. A run that cannot distinguish practical equivalence from inadequate precision is unresolved. The application is a test of the architecture's operational distinguishability, not a demonstration of machine consciousness.

8. Experiments and discriminating predictions

Experiment 1: construct the macrostate from controlled local changes

Predefine τ, perturb a local subset, and measure Δ_{BU}. Compare a nonlinear receiver-sensitive model, a linear sum, a firing-rate or activity average, and a flat microstate predictor. Bottom-up construction is supported when the perturbation changes the macro outcome through the predicted path and the effect replicates on held-out conditions.

Experiment 2: change organization while matching amount

Match total activity, number of messages, or flash count while altering phase relation, peer topology, or inhibitory organization. Equation (10) predicts that effective drive and outcome can differ despite matched amount. A null result would narrow the role assigned to organization in that preparation.

Experiment 3: physically implement and perturb context

Use a rule cue, thalamic control population, boundary controller, or shared-context register whose lower-level implementation is explicit. Match starting state and common input, then estimate Equation (12). Compare the result with a model in which the macro label is only a readout of local state. A genuine constraint model must predict intervention effects that the readout-only model misses.

Experiment 4: separate lateral influence from common drive

Manipulate peer links while preserving shared input. Use directed delays, sign changes, or targeted occlusion and compare actual pairs with shuffled or yoked controls. Lateral coordination is supported when Equation (13) survives shared-input controls and predicts a distinct mediator.

Experiment 5: test micro–macro commutation

For each allowed intervention, compare the pushed-forward low-level outcome with the corresponding high-level prediction using Equations (15) and (16). Repeat across plausible τ maps. Report failed mappings as evidence that the abstraction is incomplete for that intervention family.

Experiment 6: track role migration during learning

Estimate the role vector at multiple checkpoints. SAN predicts that receiver tuning, peer topology, context use, and environmental return can redistribute causal influence. Equation (22) tests whether that migration exceeds estimation uncertainty.

Experiment 7: compare PWD-aware and inherited baselines

Fit matched models using: rate only; band power only; phase only; generic prediction error; communication subspace; graph messages; and the receiver-relative PWD feature set. The PWD-aware model must add held-out information and survive targeted perturbation. Terminological novelty without incremental prediction or control does not meet this burden.

Experiment 8: test realization-mixture and operation-order sensitivity

For each proposed macro intervention, enumerate or sample its allowed low-level fiber, estimate Equation (25), and repeat the analysis under prespecified realization distributions rather than one unreported average. Then compute Equation (26) for nuisance families such as shared input, total activity, boundary condition, or unobserved peer state. A macro claim is stable only when its effect remains within tolerance across the admissible realization mixtures and the declared analytical order, or when any dependence is made part of the model rather than hidden.

Experiment 9: recover assigned and realized effects under interference

Randomize intervention assignment in blocks compatible with the declared exposure mapping, then record the realized dose, each recipient's peer exposure, fidelity, leakage, and attrition. Estimate the intention-to-treat contrast in Equation (28) first. Estimate exposure-specific or mediator-specific effects only where Equation (30) has adequate support. Compare the declared exposure model against no-interference, nearest-neighbor, community, and unrestricted-spillover sensitivity models. A role effect is not identified if reasonable exposure mappings reverse it and the data cannot distinguish among them.

Experiment 10: prospective power, equivalence, and capacity matching

Before the held-out run, simulate the complete analysis under the twelve countermodels in Section 5.12 and under a SAN data-generating model across weak, minimum-relevant, and stronger effects. Choose the sample size to control the primary family while keeping Type-S error and Type-M exaggeration within preregistered bounds. Freeze the smallest effect of interest, practical abstraction tolerances, capacity curve, tuning budget, and decision rule. The final comparison reports support, practical equivalence, adverse/reversal, and unresolved results for every primary estimand and comparator; no row is omitted because it is unfavorable.

9. Controlled development application

9.1 What the application can and cannot establish

Controlled Reference Study implements a controlled artificial reference system to test whether the causal grammar in Sections 5 and 8 behaves correctly when the data-generating truth is known. The application is deliberately narrower than an empirical test. Recovering an implanted role effect shows that the estimator, quality gate, and disposition logic can detect that effect in this finite design. It does not show that the same effect exists in a brain, animal collective, or deployed agent system. Likewise, recovering a true null or reversal validates failure detection rather than establishing a general absence.

The freeze preceded every Controlled Reference Study output. It fixed the role assignments, exposure mappings, separately randomized factorials, seven primary endpoints, practical bounds, multiplicity rule, model families, capacity tiers, and adverse-result policy. The training and calibration splits each contain 2,048 independent environment blocks; the development split contains 4,096. Each environment contains 32 units in four eight-unit communities. The future test contains 8,192 committed blocks, remains unauthorized, and was not generated or inspected.

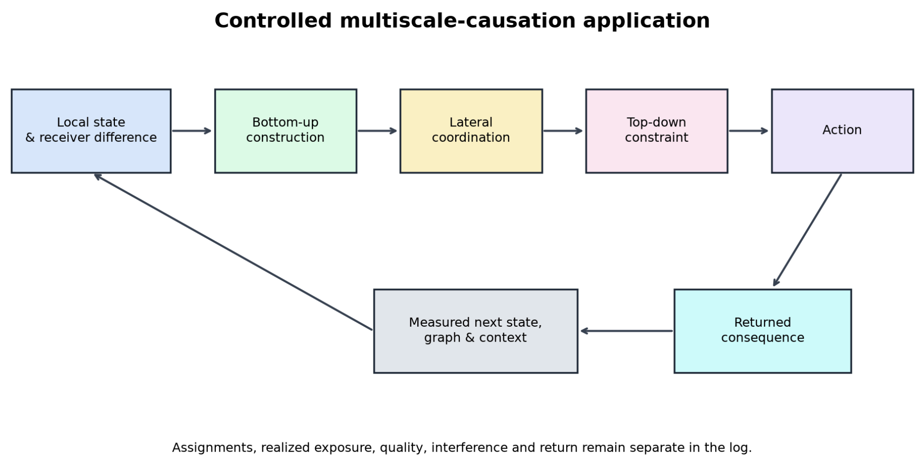
The paired development effect for role k is scaled by the control outcome standard deviation measured on the training split:

$$\widehat{d}_k = \frac{1}{n_{\text{dev}}}\sum_{b=1}^{n_{\text{dev}}}\frac{O_{b,k}(W=1)-O_{b,k}(W=0)}{s_{k,\text{train}}}. \tag{40}$$

The paired fixture removes irrelevant baseline variation while retaining independent arm noise. It is a development benchmark, not a claim that both potential outcomes are observable in an empirical unit. Bonferroni intervals cover the family of seven primary endpoints. A favorable signed effect is eligible for support only when implementation quality also passes:

$$Q_k = \mathbf{1}[L(F_k)\geq 0.90,\ U(\Lambda_k)\leq 0.10,\ \eta_k\geq 0.05], \tag{41}$$

where F_k is fidelity, Λ_k is leakage, and η_k is the minimum supported exposure probability. When $Q_k=0$, a favorable point estimate is unresolved rather than support.

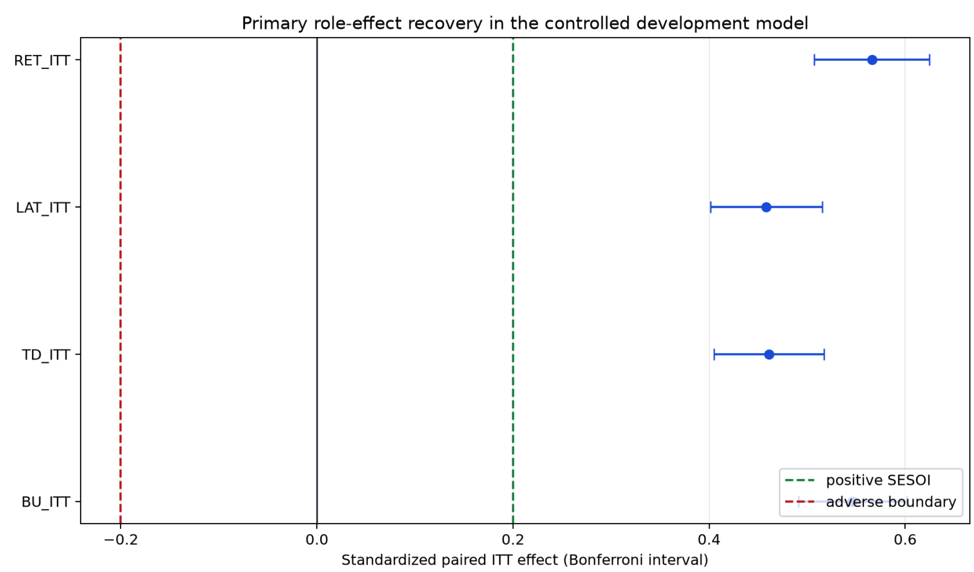


Controlled application architecture

9.2 Primary role-effect recovery

The joined-support data-generating condition implanted positive role effects while preserving distinct local state, receiver expectation, common input, rule state, assignment, realization, fidelity, leakage, and endpoint noise. The estimator recovered every role beyond the frozen 0.20-SD consequence threshold:

Role	Standardized effect	Simultaneous interval	Decision
Bottom-up construction	0.5462	[0.4912, 0.6013]	support
Top-down constraint	0.4611	[0.4049, 0.5173]	support
Lateral coordination	0.4584	[0.4016, 0.5151]	support
Returned consequence	0.5658	[0.5070, 0.6247]	support



Primary role effects

This favorable row is only one part of the test. The same estimator classified all four zero-effect fixtures as practical equivalence: their simultaneous intervals lay completely inside ([-0.20,0.20]). It classified all four sign-reversed fixtures as adverse or reversal, with intervals entirely below (-0.20). In a separate top-down fixture, the standardized effect was 0.4696 with interval [0.4157, 0.5235], but the leakage upper bound was 0.1811. Equation (41) therefore changed the disposition to unresolved. The result demonstrates that an attractive endpoint cannot override a failed role-assignment gate.

The returned-consequence test also included a no-return recurrence alternative. When the return pathway had no implanted causal effect, the estimate was 0.0449 with interval [−0.00985, 0.0996], practical equivalence at the tested horizon. The readout-only macrostate alternative similarly produced a top-down estimate of −0.0133 with interval

[−0.0684, 0.0417]. These outcomes show that recurrence and predictive readout are not silently counted as return or constraint.

9.3 Interference and shared input

The interference fixture randomized unit assignments inside four communities. Recipient outcome depended on own assignment, the fraction of treated peers in the declared community, a measured pre-intervention common input, and noise. For exposure map g, the development regression was used as a diagnostic rather than as a substitute for randomization:

$$O_{ib} = \beta_0 + \beta_W W_{ib} + \beta_G g_i(W_b) + \beta_C C_b + \varepsilon_{ib}. \tag{42}$$

The declared community map recovered a peer-exposure coefficient of 0.3592 for a true value of 0.35 and achieved the lowest mean squared error, 0.4226. Own-assignment-only and nearest-neighbor maps yielded coefficients of 0.1763 and 0.0913. The global-spillover coefficient was 0.4001 but had worse prediction error, illustrating that coefficient magnitude alone cannot validate an exposure mapping.

A known-null shared-input countermodel made peer activity and outcome covary through the same pre-intervention driver. A naive regression produced a peer coefficient of 0.6301. Once peer assignment was randomized and the prespecified pre-intervention driver was included, the estimated peer-exposure coefficient was −0.00658 around a true value of zero. No post-treatment activity or message load was inserted into a baseline adjustment set.

9.4 Amount, organization, and mediation

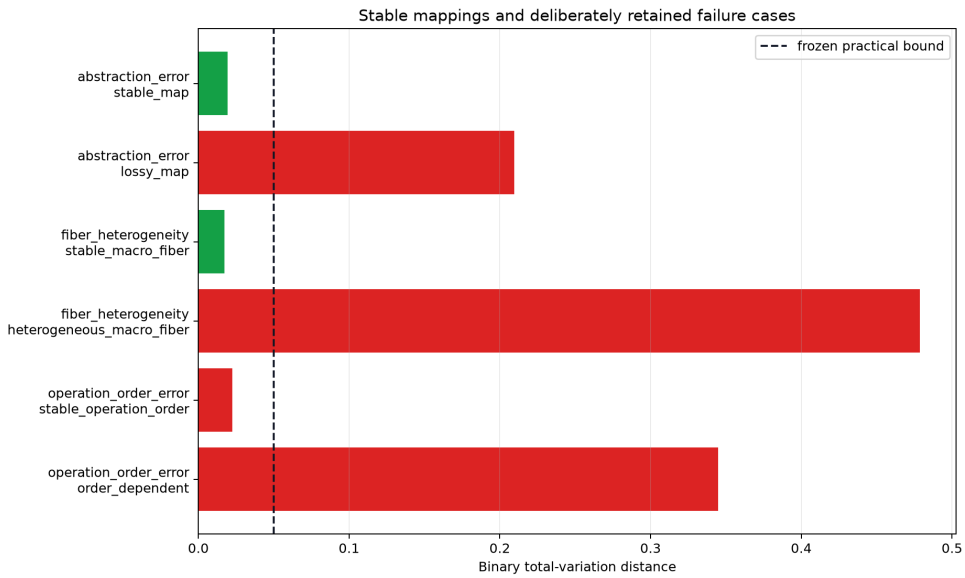
The amount-versus-organization fixture used a separately randomized 2×2 factorial. Activity dose had an effect of 0.3873 with simultaneous interval [0.3559, 0.4187], while organization had an effect of −0.0144 with interval [−0.0459, 0.0171], practical equivalence. This is an intentionally adverse result for an organization-specific explanation: when amount alone generates the outcome, the analysis does not manufacture a residual organization effect by conditioning on a post-treatment variable.

A second 2×2 design randomized both assignment and the mediator. The interventional direct effect was 0.2701 [0.2270, 0.3131], and the interventional indirect effect was 0.4196 [0.3774, 0.4618]. Because the mediator was separately randomized, these rows provide an experimental benchmark for Equations (31) and (32). They do not license a natural-effect interpretation when a future mediator is observed rather than assigned.

9.5 Abstraction, macro fibers, and operation order

The categorical branch of Equation (34) was tested using binary total variation. Stable and deliberately failed conditions were both retained:

Diagnostic	Condition	Estimate	Simultaneous interval	Decision
abstraction error	stable map	0.0193	[0, 0.0483]	practical equivalence
abstraction error	lossy map	0.2097	[0.1808, 0.2387]	adverse/fail
fiber heterogeneity	stable fiber	0.0173	[0, 0.0465]	practical equivalence
fiber heterogeneity	heterogeneous fiber	0.4788	[0.4530, 0.5045]	adverse/fail
operation-order error	nominally stable order	0.0227	[0, 0.05189]	unresolved
operation-order error	order-dependent	0.3447	[0.3168, 0.3726]	adverse/fail



Abstraction and retained failure cases

The nominally stable operation-order point estimate was below 0.05, but its simultaneous upper bound was 0.05189. The application therefore did not pass all seven primary endpoints. Calling that row equivalent would commit exactly the underpowered-null error that the four-way contract was designed to prevent. The result remains unresolved and advances as such.

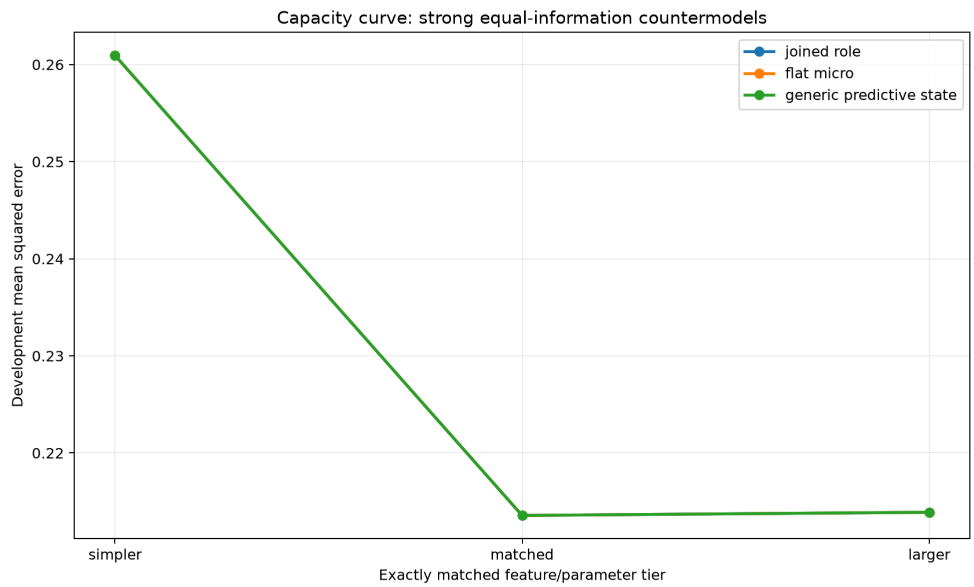
9.6 Exact-capacity countermodels

The prediction test compared a joined-role encoding, a flat-micro encoding, and a generic predictive-state encoding at 8-, 12-, and 18-feature tiers. Each family received the same underlying observations, splits, outcome rows, ridge grid, and tuning budget. The flat and generic representations used fixed orthogonal transforms of the same information, which makes them deliberately strong rather than semantically impoverished comparators. With an intercept, the matched tier contained exactly 13 parameters for every family, so the relative parameter mismatch was zero.

For model m, the matched comparison is the paired development loss difference:

$$\Delta_m = \frac{1}{n}\sum_{i=1}^{n}\Big[(O_i - \widehat{O}_{i,\text{joined}})^2 - (O_i - \widehat{O}_{i,m})^2\Big]. \tag{43}$$

The matched joined-role mean squared error was 0.2135642. The flat and generic encodings produced the same value to numerical precision; the joined-minus-best-comparator estimate was 2.17×10^{-16}, with interval [-8.86×10^{-16}, 1.32×10^{-15}]. This lies inside the frozen ±0.01 MSE equivalence region. The result is practical equivalence, not a SAN prediction win. Named roles can make interventions interpretable without automatically improving prediction over an invertible equal-information representation. A distinct predictive claim requires a new, predeclared generalization or intervention test.



Capacity curves

9.7 Reproducibility and exact boundary

Twenty-three deterministic tests passed. They cover the four decision classes, quality override, held-out prohibition, exact generator replay, support/null/reversal recovery, shared-input disambiguation, declared-map recovery, factorial integrity, mediation design, exact parameter matching, capacity-rival retention, and stable-versus-failed abstraction rows. Two complete executions produced 15 of 15 byte-identical files, including all four figures. The complete result ledger contains 22 rows and keeps every adverse, null, unresolved, and capacity-equivalent outcome.

This is a development result. The 8,192-block final namespace remains sealed. Ordered Wasserstein-1 and multivariate characteristic-MMD branches, joint-role interactions, full ADEMP coverage, randomization-based final intervals, and at least one externally generated system remain open. Those are scientific next gates, not reasons to reinterpret Controlled Reference Study's mixed outcome as a complete pass.

10. Independent replication and structural-transfer discrimination

10.1 Why the independent replication is not a post hoc repair

Controlled Reference Study produced two results that required sharper tests. Its nominally stable operation-order estimate was below the practical bound, but the simultaneous upper interval exceeded the bound by 0.00189; the correct disposition was unresolved. Its exactly matched predictor families were invertible encodings of the same information and therefore tied. Independent Replication Study preserves both results. It does not relax the 0.05 bound, retune the Controlled Reference Study generator, reuse its seed namespace, or remove its rivals.

Before any Independent Replication Study outcome, a new contract fixed an independently implemented generator, 16,384 operation-order observations per distribution, all 24 permutations of four physical channels into four causal roles, disjoint train/calibration/development partitions, three primary decisions, four model families, the ridge grid, and resource accounting. The eight development permutations were absent from both training and calibration. The parent 8,192-block final namespace remained unauthorized.

10.2 Independent operation-order replication

The operation-order replication retained binary total variation and the 0.05 practical bound. The stable fixture used independently generated Bernoulli distributions whose planning difference was bounded at 0.01. The new sample size was justified prospectively using the familywise critical value and 0.90 target power rather than selected after its output.

Controlled Reference Study remains unresolved at estimate 0.02271 and upper bound 0.05189. Independent Replication Study's new stable replication produced probabilities 0.60010 and 0.59930, total variation 0.000793, and simultaneous interval [0, 0.013753]. This is practical equivalence under the unchanged rule. The order-dependent positive control produced total variation 0.34448 [0.33208, 0.35688], adverse/fail.

Independent operation-order replication

The new row resolves the measurement question for this separately generated fixture. It does not retroactively change Controlled Reference Study, and it does not show that abstraction and marginalization commute in an external system. That requires an independently specified target process rather than probabilities set by the development generator.

10.3 The unseen-permutation problem

Suppose four physical channels carry four distinct causal roles, but hardware, anatomy, routing, or deployment changes which channel realizes which role. A predictor trained only on fixed channel identity can fail even when it receives a complete descriptor of the new mapping, because using that descriptor requires the correct channel-map interaction. A structural model can instead canonicalize the channels before prediction.

Let x_j be physical channel j, and let $P_{jr} = 1$ when physical channel j realizes role r. The canonical role coordinate is

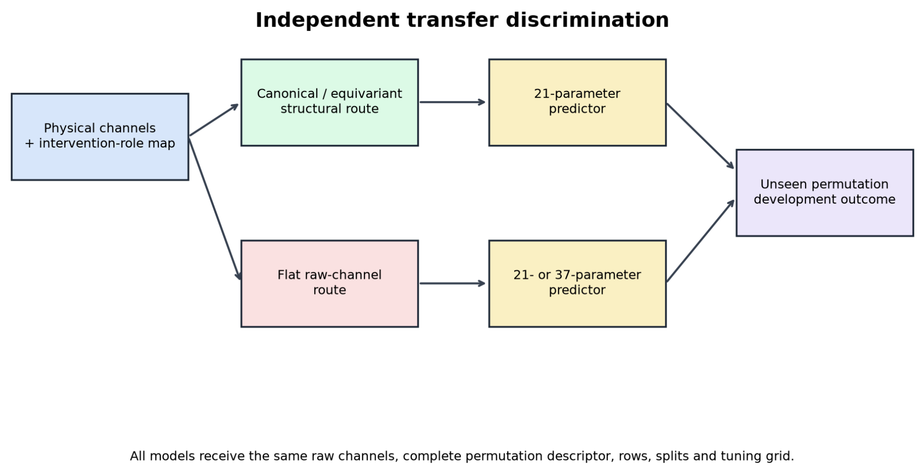
$$c_r = \sum_{j=1}^{4} P_{jr} x_j. \tag{44}$$

The map is an intervention-role assignment, not a learned semantic label. A role-aware or role-name-free group-equivariant predictor can use the same operation. Under a physical-channel permutation (Pi), a correct canonicalization satisfies

$$c(\Pi x, \Pi P) = c(x, P). \tag{45}$$

The generator assigned signed heterogeneous predictive loads to the four canonical coordinates. Those loads make remapping consequential but do not assert that bottom-up, top-down, lateral, or returned-consequence effects have universal positive or negative signs.

All 24 permutations were used exactly once across the three partitions: eight in training, eight in calibration, and eight in development. Each model received the same physical channels, complete 16-entry permutation descriptor, outcomes, rows, partitions, four ridge choices, and tuning trials.



Independent transfer architecture

10.4 Four deliberately strong models

The primary structural model used four canonical coordinates plus the complete descriptor, for 20 features and 21 fitted parameters including the intercept. The generic group-equivariant model used the same canonical information under a fixed orthogonal rotation and also had 21 parameters. It is the strongest specificity control: if it ties, the result belongs to equivariance rather than SAN terminology.

The matched flat model used four raw physical channels plus the same 16-entry descriptor, also 20 features and 21 parameters. It had access to the mapping but no fixed products between channel value and mapping entry. The larger flat model added all 16 channel-descriptor products. It therefore had 36 features and 37 parameters and could express the canonicalization without using named roles.

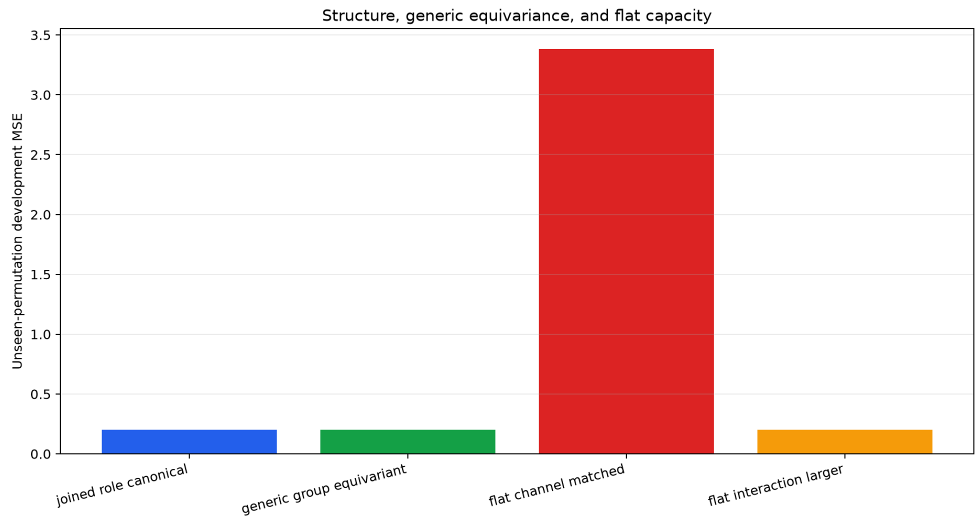
The primary transfer contrast was the paired development loss difference

$$\Delta_{\text{flat}} = \frac{1}{n}\sum_{i=1}^{n}\left[(O_i - \widehat{O}_{i,\text{flat}})^2 - (O_i - \widehat{O}_{i,\text{joined}})^2\right]. \tag{46}$$

Structural-transfer support required the simultaneous lower bound to exceed 0.05 MSE. SAN-specific nonuniqueness was tested separately by requiring the full joined-minus-generic interval to lie within ±0.005 MSE.

10.5 Transfer results

Model	Parameters	Development MSE	Result role
joined-role canonical	21	0.200045	primary structural model
generic group-equivariant	21	0.200045	strongest specificity rival
flat channel matched	21	3.382506	equal-parameter raw-channel rival
flat interaction larger	37	0.200469	higher-capacity flat rival



Unseen-permutation model comparison

The matched flat-minus-joined MSE contrast was 3.18246 with simultaneous interval [3.05369, 3.31123], exceeding the 0.05 decision margin. A correct structural map therefore supplied a decisive transfer benefit at the same trainable parameter count in this fixture.

That is not a SAN-specific win. The joined-minus-generic estimate was 3.70×10^{-15}, with interval [-1.57×10^{-14}, 2.31×10^{-14}], practical equivalence. The generic equivariant architecture realizes the same invariant without SAN role names.

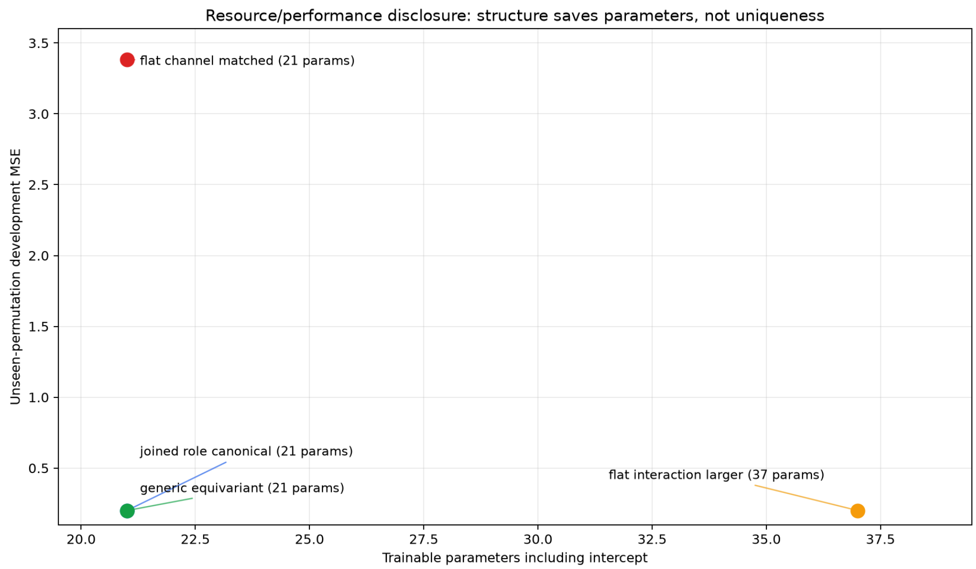
Nor is the structured model uniquely expressive. The 37-parameter flat interaction model differed from the joined model by 0.000424 MSE [−0.000194, 0.001042], practical equivalence inside ±0.005. It catches up when given the interaction capacity required to multiply the physical channels by their descriptor.

10.6 Resource interpretation

Let $p_s = 21$ be the structured parameter count and $p_f = 37$ the smallest tested flat interaction count that recovers equivalence. The disclosed parameter ratio is

$$\rho_p = \frac{p_f}{p_s} = \frac{37}{21} \approx 1.762. \tag{47}$$

The resource ledger also records preprocessing multiplications per row, serialized float-64 weight bytes, and a deterministic normal-equation fit-operation proxy. The 21-parameter models use 168 weight bytes and a fit proxy of 3,615,759 operations; the 37-parameter model uses 296 bytes and a proxy of 11,231,732. These are analytical disclosures, not measured hardware energy or wall-clock claims. The generic equivariant model also spends more fixed preprocessing multiplications than the named canonical implementation while retaining the same fitted parameter count.



Resource and performance disclosure

The defensible conclusion is therefore precise: a correct structural mapping reduced trainable capacity required for transfer across unseen permutations. The result does not establish that SAN names are the only way to encode the invariant, that flat models cannot express it, or that the same efficiency occurs in a biological system.

10.7 Reproducibility and remaining boundary

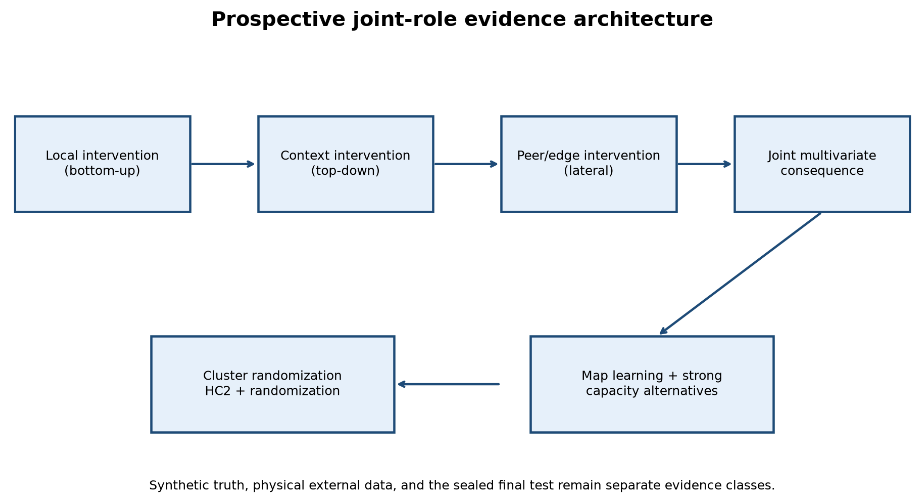
Sixteen deterministic tests passed. They verify partition completeness, permutation disjointness, held-out rejection, exact generator replay, canonical reconstruction, feature and parameter counts, independent operation-order sample size, stable equivalence, failed positive control, matched-flat discrimination, generic-equivariant nonuniqueness, larger-flat recovery, Controlled Reference Study preservation, and overwrite refusal. Two complete executions produced nine of nine byte-identical result and figure files.

All three Independent Replication Study primary questions passed their frozen decision rule, but the larger Paper 9 program remains unfinished. The map was supplied rather than learned; the permutation family is finite and bijective; resource figures are analytical; the generator is artificial; ordered and multivariate outcomes, joint roles, the full ADEMP grid, formal invariants, and external data remain open. The parent final test stays sealed.

11. Joint-role estimation, learned maps, and a physical transfer check

Independent Replication Study established that a supplied structural map can reduce the fitted capacity required to transfer across unseen channel permutations. That result did not answer how the three roles behave when they are changed together, whether interval procedures retain their advertised behavior across difficult data-generating mechanisms, whether the map can be learned rather than supplied, or whether any bounded part of the analysis can run on measurements from a physical system. Joint-Role ADEMP Study treats those as four different questions. It does not combine them into one success score.

Before any Joint-Role ADEMP Study output existed, the contract fixed the data-generating mechanisms, seed namespace, estimands, comparison methods, performance measures, practical bounds, randomization count, map-learning conditions, model families, data partitions, physical archive hash, and claim boundaries. The source-through-replication artifacts were hash-locked. The earlier Controlled Reference Study unresolved operation-order row, the Controlled Reference Study equal-capacity ties, the Independent Replication Study generic-equivariant tie, the Independent Replication Study larger-flat recovery, and the absence of a SAN-specific predictive increment were declared mandatory readback items. The separately committed final 8,192-block test remained unauthorized.



Joint-role evidence architecture

11.1 Joint bottom-up, top-down, and lateral assignments

The new synthetic experiment randomizes three binary factors together. Each of eight blocks contains the complete eight-cell factorial; the randomized unit is the cluster, and each cluster contains twelve lower-level observations whose mean supplies the analyzed outcome. Effect coding uses -1 and $+1$, so all main effects and interactions are estimable in one orthogonal design:

$$\begin{aligned} Y_{bc} = \beta_0 + \beta_B B_{bc} + \beta_T T_{bc} + \beta_L L_{bc} + \beta_{BT} B_{bc} T_{bc} + \beta_{BL} B_{bc} L_{bc} \\ + \beta_{TL} T_{bc} L_{bc} + \beta_{BTL} B_{bc} T_{bc} L_{bc} + u_b + \epsilon_{bc}, \end{aligned} \tag{48}$$

where b indexes blocks and c indexes factorial cells. B, T, and L are experimental assignments to the bottom-up, top-down, and lateral roles in this controlled system; they are not labels inferred from an observational correlation. The interaction terms ask whether the consequence of changing one role depends on the joint state of another role. A nonzero interaction is therefore not a new fourth kind of cause. It is evidence that the declared causal roles are not additively separable in that preparation.

Uncertainty is computed at the randomized cluster unit. With design matrix X, residual $\hat{e}_i$, and leverage h_i, the HC2 sandwich used in the application is

$$\widehat{\mathrm{Var}}_{\mathrm{HC2}}(\hat{\beta}) = (X^\top X)^{-1} X^\top \operatorname{diag}\left(\frac{\hat{e}_i^2}{1-h_i}\right) X (X^\top X)^{-1}. \tag{49}$$

The seven primary coefficients use a Bonferroni simultaneous normal critical value of 2.69011. HC2 is a finite-sample leverage correction, not a guarantee that every cluster design is well approximated by a normal interval [50]. Joint-Role ADEMP Study therefore evaluates the procedure itself across repeated data-generating mechanisms and retains a design-based randomization diagnostic rather than treating one standard-error formula as self-validating.

11.2 ADEMP: evaluating the measuring procedure before using it as an oracle

Morris, White, and Crowther organize simulation studies around aims, data-generating mechanisms, estimands, methods, and performance measures—ADEMP [47]. Joint-Role ADEMP Study uses 10,240 independent replicates in each of seven scenarios: null effects, minimum effects, strong effects, high intracluster correlation, outcome-dependent attrition, nonlinear misspecification, and shared input. It compares the declared cluster-HC2 analysis with a deliberately incorrect unit-precision calculation and a main-effects-only misspecification. Every coefficient and scenario remains in the 119-row ledger.

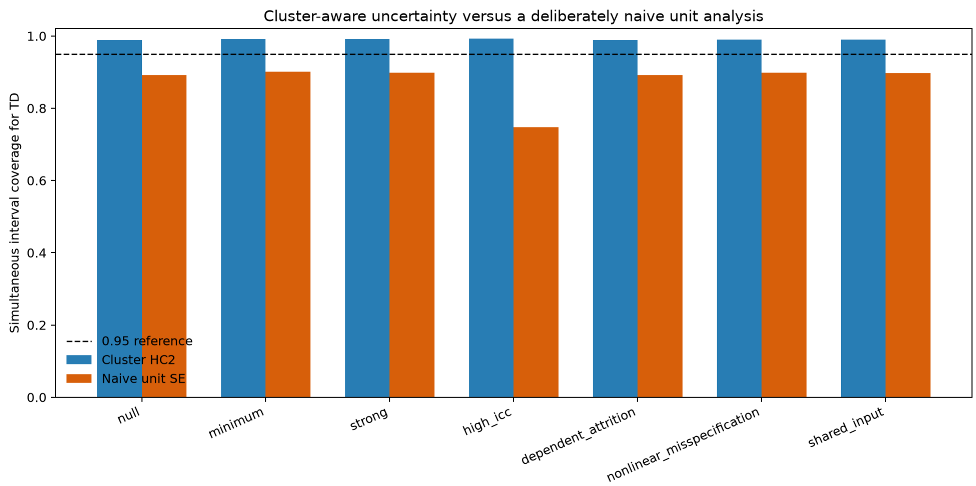
For a binomial performance estimate $\hat{q}$ from R replicates, the reported Monte Carlo standard error is

$$\mathrm{MCSE}(\hat{q}) = \sqrt{\frac{\hat{q}(1-\hat{q})}{R}}. \tag{50}$$

The largest reported binomial MCSE was 0.004937, below the frozen 0.005 ceiling. The cluster-HC2 intervals produced per-endpoint simultaneous coverage between approximately 0.988 and 0.992 throughout the seven tested mechanisms. Their conservatism is expected because the seven-coefficient family uses a Bonferroni critical value. The naive unit-precision analysis preserved the same point estimates but understated their standard errors: its coverage ranged from about 0.742 in the high-ICC scenario to 0.900 in the easier scenarios.

Scenario and coefficient	Method	Truth	Bias	Coverage	Rejection or power
null, BU	cluster HC2	0.00	−0.00058	0.99199	0.00801
null, BU	naive unit precision	0.00	−0.00058	0.89893	0.10107

Scenario and coefficient	Method	Truth	Bias	Coverage	Rejection or power
minimum, BU	cluster HC2	0.25	0.00047	0.99004	0.93506
minimum, $B \times T \times L$	cluster HC2	0.04	−0.00025	0.99150	0.02695
high ICC, BU	cluster HC2	0.25	0.00010	0.99023	0.61396
high ICC, BU	naive unit precision	0.25	0.00010	0.74990	0.96621
dependent attrition, TD	cluster HC2	0.20	$< 10^{-8}$	0.98857	0.67927



ADEMP cluster-aware coverage

The minimum three-way effect shows why a complete factorial is not the same as an adequately powered one. Its true coefficient was 0.04, but simultaneous power was only 0.02695; among the small set of significant estimates, Type-M exaggeration was 4.27 and Type-S error was 0.0181. The correct conclusion is that the design can represent the interaction but cannot reliably detect an interaction that small at the chosen familywise threshold. The ADEMP grid therefore supplies both a working region and a failure map.

11.3 Assignment-based randomization evidence

One strong-effect realization was additionally analyzed by permuting complete factorial cell assignments within each of the eight blocks. For observed statistic T_k^{obs}, $R = 4096$ assignment draws, and the maximum absolute statistic $M^{(r)}$ across seven effects, the familywise diagnostic is

$$\widehat{p}_{k,\max T} = \frac{1 + \sum_{r=1}^{R} \mathbf{1}\left[M^{(r)} \geq |T_k^{\text{obs}}|\right]}{R+1}. \tag{51}$$

Effect	Observed $\lvert t \rvert$	Max-T p	Joint-Role ADEMP Study disposition
BU	9.570	0.000244	detected in this realization
TD	7.682	0.000244	detected in this realization
LAT	4.518	0.000976	detected in this realization
BU×TD	2.582	0.13010	not familywise detected
BU×LAT	2.888	0.06663	not familywise detected
TD×LAT	1.844	0.46912	not familywise detected
BU×TD×LAT	1.081	0.91286	not familywise detected

The three main effects pass the 0.05 familywise threshold; none of the four planted interactions does. That mixed result remains visible. It agrees with the ADEMP finding that small interactions can be weakly powered even in a complete factorial.

This permutation is a design-based maximum-statistic diagnostic of assignment sensitivity. It is not promoted into an exact isolated test of every non-sharp coefficient null when other role effects are present. Exact randomization claims under interference or composite nulls require an artificial experiment or conditioning argument suited to the null being tested [51]. A later confirmatory design should therefore predeclare contrast-specific randomization hypotheses, not merely permute all labels and inherit the word “exact.”

11.4 Ordered and multivariate outcomes

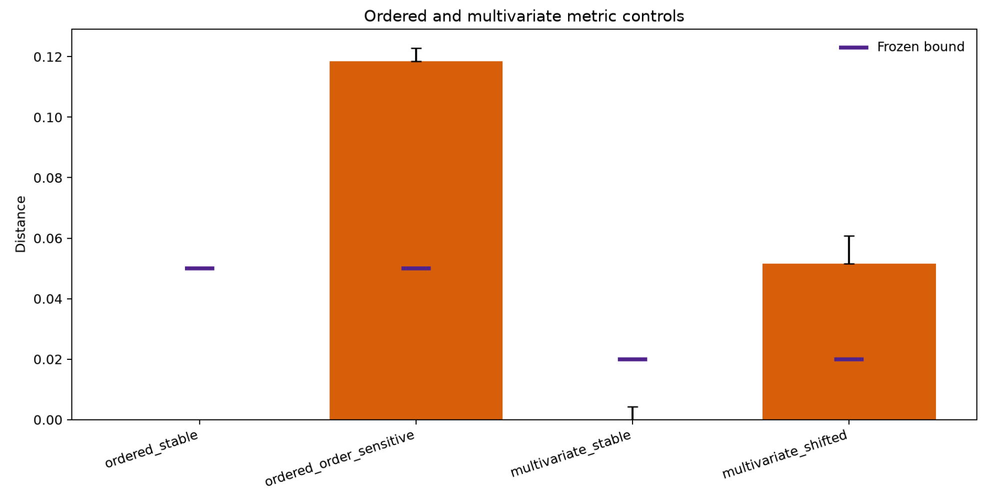
Prospective Analysis Plan declared different distances for outcomes with different structure. Joint-Role ADEMP Study finally executes the ordered and multivariate branches. For ordered categories $0, \ldots, K-1$, the normalized discrete 1-Wasserstein distance is

$$W_1^{\text{norm}}(P,Q) = \frac{1}{K-1}\sum_{k=0}^{K-2}|F_P(k) - F_Q(k)|. \tag{52}$$

This preserves category order rather than treating "one step apart" and "four steps apart" as the same mismatch [52]. For multivariate samples, the Gaussian-kernel maximum mean discrepancy is

$$\widehat{\text{MMD}}_\kappa^2 = \frac{1}{n(n-1)}\sum_{i\neq j}\kappa(x_i,x_j) + \frac{1}{m(m-1)}\sum_{i\neq j}\kappa(y_i,y_j) - \frac{2}{nm}\sum_{i,j}\kappa(x_i,y_j), \tag{53}$$

with bandwidth fixed from calibration data by the median pairwise squared distance [53]. The exact-stable ordered fixture had distance 0 with interval [0, 0], inside the 0.05 bound. The order-sensitive fixture was 0.11847 [0.11399, 0.12289], above the same bound. The exact point MMD for identical multivariate samples was 0; its random-feature bootstrap interval [0.00041, 0.00422] remained inside the 0.02 bound. The shifted fixture was 0.05167 [0.03571, 0.06072], above that bound. Both stable and deliberately changed controls were therefore classified correctly without exchanging metrics or tolerances after outcomes.



Ordered and multivariate controls

11.5 Learning the role map and exposing its failure surface

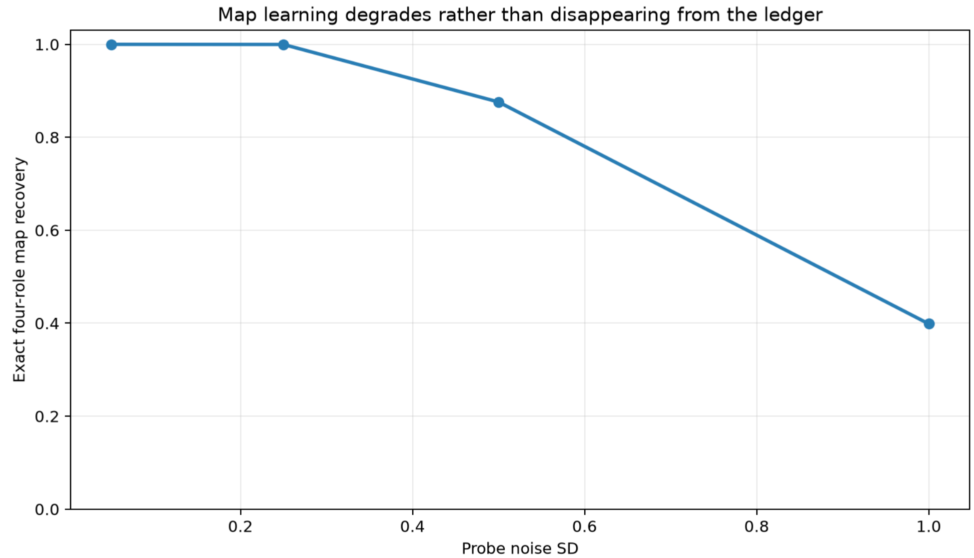
Independent Replication Study supplied the map from four physical channels to bottom-up, top-down, lateral, and returned-consequence coordinates. Joint-Role ADEMP Study replaces that gift with four noisy probes. For probe matrix Q, the learner searches all $4! = 24$ bijections and selects

$$\widehat{\pi} = \arg\max_{\pi\in S_4}\sum_{j=1}^{4} Q_{j,\pi(j)}, \tag{54}$$

with a frozen lexicographic tie break. This is an intentionally transparent finite learner. It tests whether a declared structural map can be recovered from intervention-like probes; it does not solve general scientific role discovery.

Probe condition	Exact-map accuracy	Channel-label accuracy	Canonicalization MSE
noise 0.05, complete	1.0000	1.0000	0.0000
noise 0.25, complete	0.99976	0.99988	0.000015
noise 0.50, complete	0.87598	0.93134	0.13729
noise 1.00, complete	0.39868	0.61743	0.74720
noise 0.05, 25% probes missing	0.84888	0.91943	0.17244
noise 1.00, 25% probes missing	0.28979	0.53674	0.91368
deliberately non-bijective	0.0000	0.7500	not defined

Post-learning corruption produced the same graded failure: exact accuracy was 0.90186 at 10% corruption, 0.74634 at 25%, and 0.49146 at 50%. The non-bijective control is more important than a low score. It shows that a one-to-one permutation model cannot represent two physical channels assigned to the same role. The correct response is a different map class—such as a many-to-one stochastic or set-valued transport—not a favorable reinterpretation of the bijection.



Role-map learning stress

11.6 Strong capacity and symmetry alternatives

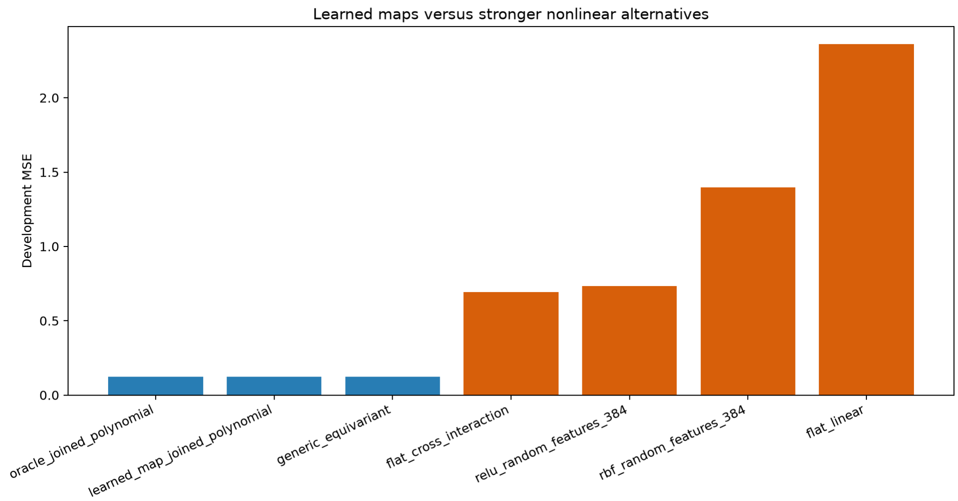
The transfer benchmark uses 6,144 training, 2,048 calibration, and 4,096 development rows. The probe noise is 0.25, yielding development exact-map accuracy 0.99976. Seven frozen model families receive the same raw channels, probe information, outcome, and split except for the oracle, which is separately labeled because it receives the true map. The two 384-feature alternatives implement fixed random Fourier or ReLU features followed by a fitted ridge head [55]. The group-equivariant comparator carries the same learned canonical information through a fixed orthogonal change of basis, expressing the broader principle that symmetry-aware representations can share structure and reduce sample requirements [54].

For model f, resource reporting is not collapsed into parameter count alone:

$$\mathfrak{R}(f) = \left(p_f,\ q_f,\ 8n_{\text{train}}q_f,\ n_{\text{train}}q_f^2 + \frac{q_f^3}{3}\right), \tag{55}$$

where $p_f = q_f + 1$ is the fitted linear-head parameter count, q_f the feature count, the third component is float-64 design storage in bytes, and the fourth is a deterministic fit-operation proxy. Fixed random-feature weights are disclosed separately. These are comparable analytical quantities, not measurements of hardware energy or elapsed time.

Model	Fitted parameters	Fixed feature parameters	Development MSE
oracle joined polynomial	28	0	0.124590
learned-map joined polynomial	28	0	0.124675
generic equivariant	28	729	0.124675
flat linear	21	0	2.359785
flat cross-interaction	89	0	0.693113
RBF random features	385	8,064	1.397833
ReLU random features	385	8,064	0.733606



Strong model comparison

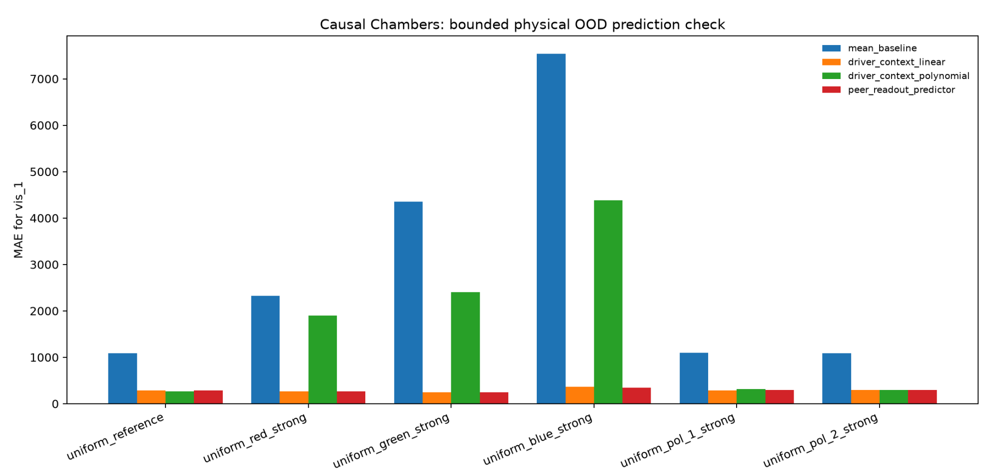
The learned joined model was practically equivalent to the oracle: oracle-minus-learned paired MSE was −0.0000853 [−0.0003105, 0.0001398], inside the frozen ±0.01 region. The generic equivariant comparator tied to numerical precision. Relative to the learned joined model, simultaneous alternative-minus-joined intervals were [2.08315, 2.38707] for flat linear, [0.50825, 0.62862] for flat cross-interaction, [1.15523, 1.39108] for RBF random features, and [0.54951, 0.66835] for ReLU random features.

This result does not erase the Independent Replication Study result in which a larger flat interaction model caught up. The generators and target functions differ. Independent Replication Study demonstrates that ordinary interaction capacity can recover its finite remapping function. Joint-Role ADEMP Study demonstrates that the particular 88-feature flat cross family and two frozen 384-feature random families did not recover the richer role-polynomial target under the declared data and tuning budget. The generic equivariant tie remains the strongest specificity result: what transferred was structural symmetry and role-aligned interaction, not ownership by SAN terminology.

11.7 A fixed physical-data route

The external route uses the exact public light-tunnel archive from Gamella, Peters, and Bühlmann's Causal Chambers project [56]. The local ZIP is fixed by SHA-256 8781960B5FFF5D752F57566393D4B0F8706CAC9B2CD2A3D3226442F17C7BAB60, the repository is pinned to commit 6da4e646cf62bc902f87911c866298a8c98ad2eb, and the data are licensed CC BY 4.0. Models are trained and calibrated on fixed partitions of the uniform-reference experiment, then evaluated without refitting on that reference and five strong interventions. The outcome is sensor vis_1.

Across the six experiments, median MAE was 1,709.91 for the reference-mean baseline, 284.23 for the driver/context linear model, 1,106.24 for the polynomial model, and 289.68 for the peer-readout predictor. The linear driver/context model was the best median performer; adding other sensor readouts was close but did not supply a general improvement, and the selected polynomial family transferred poorly. This mixed pattern is valuable because a physical archive can expose misspecification that a generator designed by the analyst may hide.



Causal Chambers out-of-distribution check

The physical archive does not label its variables as bottom-up, top-down, or lateral SAN roles. A peer sensor can improve prediction because it shares a readout or common cause; that does not make it a lateral intervention. The result is therefore bounded to out-of-distribution prediction across controlled physical interventions. It supplies a public-data route and a misspecification check, not validation of the synthetic role taxonomy, neural implementation, multiscale biology, or consciousness. The Causal Chambers authors make the same general distinction: these systems are useful physical sanity checks, while success need not transfer to larger or more complex systems [56].

11.8 Reproducibility, formal invariants, and the remaining gate

Twenty-four deterministic tests pass. They verify parent-artifact hashes, pre-outcome freeze, preservation of earlier adverse and equivalence results, complete blocked factorial construction, full rank, noiseless coefficient recovery, randomization draw count, ordered and multivariate stable controls, all 24 noiseless permutations, non-bijective exclusion, transfer-split sizes, seven comparator families, deterministic generation, exact external-archive identity, external route loading, overwrite refusal, and the sealed final-test boundary. Two clean post-amendment executions reproduce all fifteen scientific and figure files byte for byte; they also match the original two executions byte for byte. The amendment changed only warning-producing implementation branches and was required to reproduce the original outputs exactly.

A bounded Lean 4 leaf proves twelve finite invariants used by the application: the complete binary three-role factorial has eight cells; effect coding is signed and squares to one; bijective four-channel role maps preserve values in both transport directions; the four-way interval classifier is exhaustive and single-valued; and the declared joined and flat-cross feature and fitted-parameter counts are exact. The global compiler lease returned exit code 0 under Lean 4.30.0 and mathlib revision c5ea00351c28e24afc9f0f84379aa41082b1188f. The source contains no sorry, admit, custom axiom, or native-decision axiom. The formal result is intentionally finite: it does not prove estimator coverage, map recovery under noise, the scientific correctness of a recovered role map, external validity, biology, consciousness, or SAN as a complete theory.

Joint-Role ADEMP Study closes the executable ordered, multivariate, joint-role, ADEMP, learned-map, strong-alternative, physical-data, and finite-proof gates at development scale. It also exposes limits that remain decisive: small interactions were weakly powered; no planted interaction passed the one-realization familywise randomization diagnostic; a generic equivariant model tied; map recovery degraded and failed under model-class violation; the physical data do not carry the complete role labels; and the final test is still sealed. The next study phase should strengthen contrast-specific randomization inference, non-bijective map objects, and genuinely independent external role interventions before any final-test release.

12. Contrast-specific randomization, relational maps, and trained alternatives

Joint-Role ADEMP Study exposed three problems that a favorable average score could conceal. Its one-realization max-T diagnostic was not an exact test of each non-sharp factorial coefficient while other effects were present. Its map learner assumed a one-to-one permutation even after the non-bijective control showed that the map class was wrong. Its strong raw-feature alternatives included large fixed feature banks but not end-to-end-trained nonlinear networks. Contrast-and-Relation Study addresses those problems under a new prospective contract. The contract, output schema, role-intervention protocol, seeds, splits, estimands, model families, comparison bounds, and expected artifacts were frozen before the first outcome. The source-through-joint-role artifacts and the parent final-test seal were hash-locked.

12.1 Exact tests require an exact artificial experiment

Each of seven role contrasts now receives its own matched-pair artificial experiment. For pair i, assignment $Z_i \in \{0,1\}$, and potential outcomes $Y_i(1), Y_i(0)$, the pair difference is

$$D_i = Z_i\{Y_i(1) - Y_i(0)\} + (1 - Z_i)\{Y_i(0) - Y_i(1)\}. \tag{56}$$

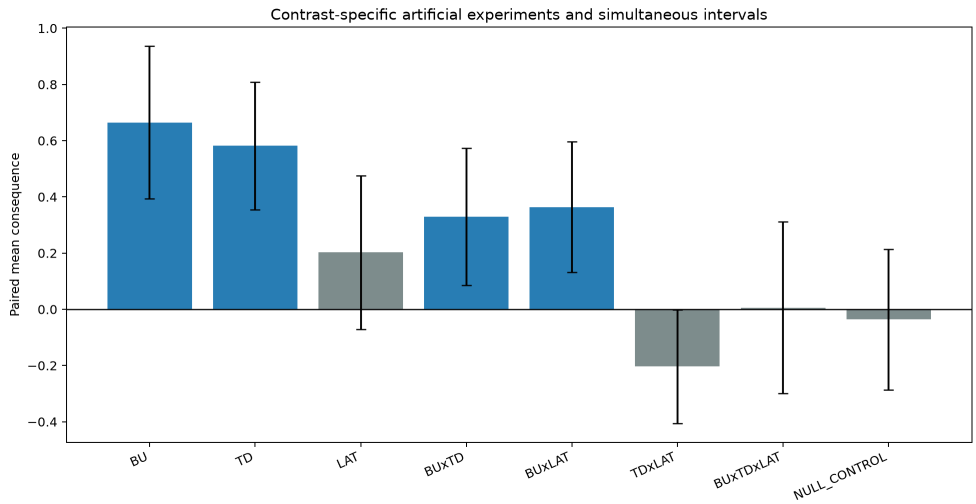
Under the isolated sharp null $H_0 : Y_i(1) = Y_i(0)$ for all sixteen pairs, every one of the $2^{16} = 65{,}536$ sign assignments is enumerated. For the absolute mean-difference statistic T, the exact two-sided probability is

$$p_{\text{exact}} = \frac{1}{2^{16}} \sum_{z \in \{0,1\}^{16}} \mathbf{1}\{|T(z)| \geq |T(z_{\text{obs}})|\}. \tag{57}$$

The seven role contrasts use Bonferroni-adjusted exact probabilities and simultaneous normal intervals; the null control is reported separately. This construction earns the word *exact* only for each isolated sharp-null experiment. It does not turn a contrast in the joint factorial into a sharp null while the other factors continue to act.

Isolated contrast	Implanted mean	Estimate	Simultaneous interval	Exact p	Bonferroni p	Disposition
BU	0.55	0.66471	[0.39298, 0.93643]	0.000061	0.000427	support
TD	0.45	0.58160	[0.35455, 0.80865]	0.000061	0.000427	support
LAT	0.35	0.20210	[−0.07197, 0.47617]	0.06494	0.45459	unresolved
BU×TD	0.25	0.32974	[0.08603, 0.57345]	0.00333	0.02328	support
BU×LAT	0.18	0.36381	[0.13178, 0.59584]	0.000519	0.00363	support
TD×LAT	−0.12	−0.20363	[−0.40596, −0.00130]	0.01813	0.12689	unresolved

Isolated contrast	Implanted mean	Estimate	Simultaneous interval	Exact	Bonferroni	Disposition
BU×TD×LAT	0.08	0.00609	[−0.29969, 0.31187]	0.95840	1.00000	unresolved
null control	0.00	−0.03648	[−0.28603, 0.21306]	0.69504	not applied	null not rejected



Contrast-specific exact randomization results

The result is intentionally mixed. Four isolated contrasts pass the familywise rule; three do not; and the true null is not rejected. In particular, a conventional interval for TD×LAT barely excludes zero while the exact multiplicity-adjusted test remains unresolved. Contrast-and-Relation Study follows the frozen joint decision rule rather than selecting the more favorable analysis.

12.2 Replace a forced permutation with a relational map

A physical channel need not instantiate exactly one scientific role, and several channels can contribute to the same role. Contrast-and-Relation Study therefore replaces the permutation with a nonnegative relation $R \in [0,1]^{6\times 4}$ whose rows are stochastic:

$$R_{cj} \geq 0, \qquad \sum_{j=1}^{4} R_{cj} = 1 \quad \text{for every observed channel } c. \tag{58}$$

Multiple rows can put mass on one role, a row can split mass across roles, and an entire role column can be absent. For physical values x_c, the mass-normalized role coordinate and its presence indicator are

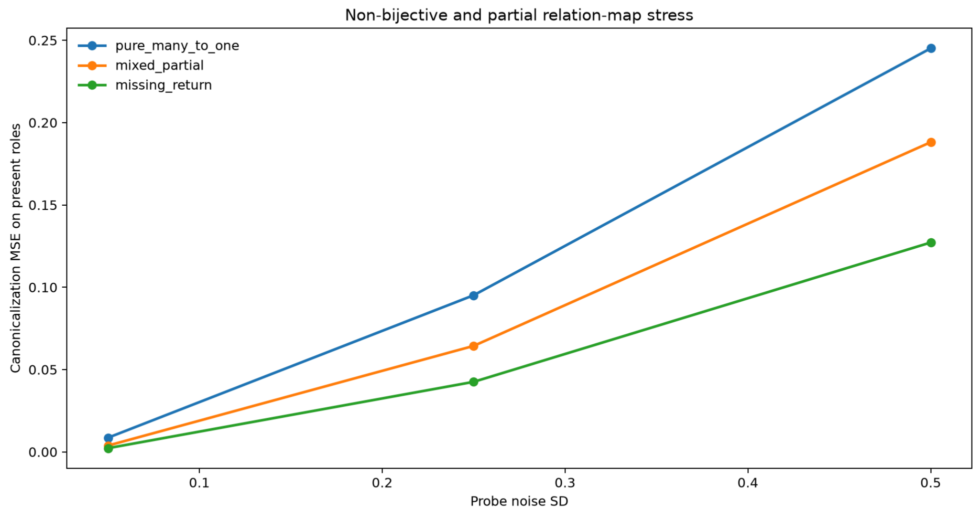
$$z_j = \begin{cases} \dfrac{\sum_c R_{cj} x_c}{\sum_c R_{cj}}, & \sum_c R_{cj} > 0, \\ 0, & \sum_c R_{cj} = 0, \end{cases} \qquad a_j = \mathbf{1}\left\{\sum_c R_{cj} > 0\right\}. \tag{59}$$

The learner rectifies noisy probe values and normalizes each observed channel row. With probe noise E, observation mask M, and row-normalization operator $\mathcal{N}_{\mathrm{row}}$,

$$\widehat{R} = \mathcal{N}_{\mathrm{row}}(M \odot \max(0, R + E)). \tag{60}$$

Missing probes are set to zero before normalization. This transparent learner is not a general solution to role discovery; its purpose is to make misspecification and information loss measurable.

Across 4,096 repetitions per condition, low-noise complete probes recovered all three relation templates closely. For pure many-to-one, mixed partial, and missing-return templates, relation MSE was 0.00199, 0.00160, and 0.00175, respectively. Missingness and probe noise caused monotone practical damage. Pure-many-to-one canonicalization MSE rose to 0.51886 at noise 0.50 with 25% missing probes. Missing-return detection fell from 0.738 at low noise with complete probes to 0.061 at noise 0.50 despite complete probes. The relation can represent an absent role; the learner is not guaranteed to discover it.



Relational-map stress and failure surface

These adverse rows are part of the result, not a reason to retune the learner. They show that expanding the model class fixes one structural impossibility while creating a harder identification problem. A future physical study must supply interventions that distinguish shared physical channels, partial participation, and genuine role absence.

12.3 Stronger trained alternatives and the cost of map error

The new transfer fixture uses 4,096 training rows, 1,024 calibration rows, and 2,048 development rows. It compares an oracle relational polynomial, a learned relational polynomial, a hard-argmax version of the learned relation, a flat quadratic ridge model, and two end-to-end-trained rectified-linear networks. For a one-hidden-layer network of width h,

$$f_h(x) = b_2 + w_2^{\top} \max(0, W_1 x + b_1). \tag{61}$$

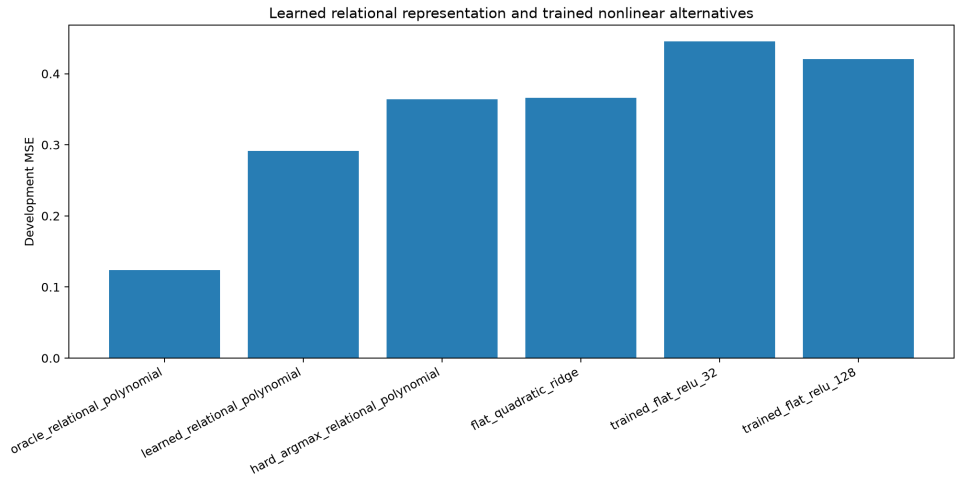
The width-32 and width-128 networks were trained for the frozen 100-epoch budget with Adam [58] across three initialization candidates, with the calibration split selecting the seed. Cybenko's universal-approximation theorem concerns representational existence under its conditions [57]. It does not guarantee that a finite network, finite sample, optimizer, seed set, or training budget will recover this target. The neural alternatives are therefore strong modern comparators, not an impossible-to-falsify appeal to asymptotic expressivity.

For every fitted model, Contrast-and-Relation Study reports a bounded resource vector

$$\mathfrak{R}_7(f) = (q_f, p_f, b_f, o_f, e_f), \tag{62}$$

where q_f is input or feature count, p_f trainable parameters, b_f float-64 weight bytes, o_f a declared training-operation proxy, and e_f epochs run. These quantities prevent a smaller error from being described without its capacity and training context.

Model	Inputs/features	Trainable parameters	Development MSE	Resource note
oracle relational polynomial	42	43	0.12364	true relation supplied
learned relational polynomial	42	43	0.29113	noisy relation learned
hard-argmax relational polynomial	42	43	0.36426	partial mass discarded
flat quadratic ridge	204	205	0.36595	raw physical/probe expansion
trained flat ReLU, width 32	30	1,025	0.44577	100 Adam epochs
trained flat ReLU, width 128	30	4,097	0.42074	100 Adam epochs



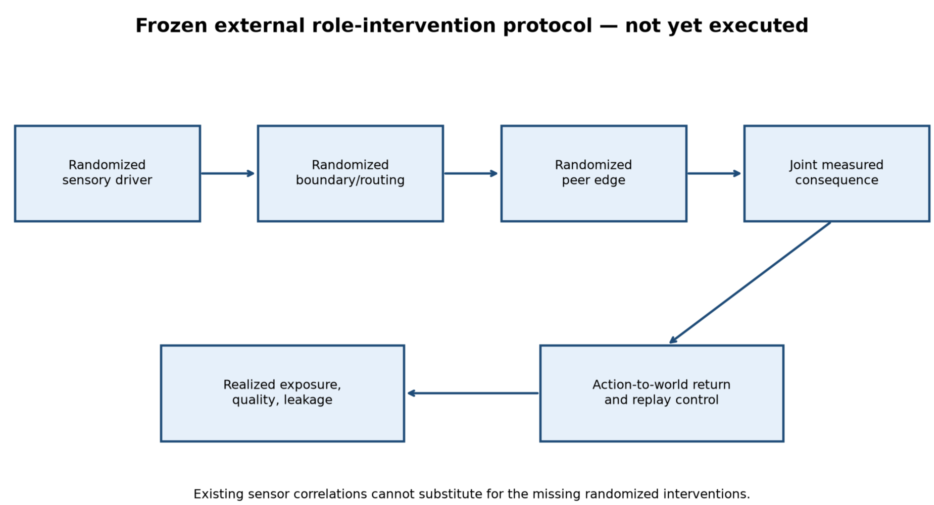
Trained and structural alternative comparison

The oracle-minus-learned paired MSE difference was −0.16748 [−0.19113, −0.14384], so the oracle was better and map-estimation error had a measurable cost. Relative to the learned relational model, alternative-minus-learned intervals were [0.03649, 0.10977] for hard argmax, [0.05122, 0.09842] for flat quadratic ridge, [0.12197, 0.18731] for width-32 ReLU, and [0.09898, 0.16024] for width-128 ReLU. All four were worse under the frozen development comparison.

This does not erase Independent Replication Study's larger-flat recovery or Joint-Role ADEMP Study's generic-equivariant tie. Those are mandatory members of the cumulative result. Nor does it show that trained nonlinear models cannot learn the target under other architectures, data scales, optimization budgets, or inductive biases. Contrast-and-Relation Study establishes only that, in this fixed finite fixture, the learned soft relation preserved useful structure that the tested hard, flat, and trained alternatives did not recover as efficiently.

12.4 The physical role experiment is specified, not simulated

The Causal Chambers archive used in Joint-Role ADEMP Study supports a physical out-of-distribution prediction check, but it does not identify all four roles. Contrast-and-Relation Study therefore freezes a prospective protocol requiring independently randomized sensory-driver, boundary/routing, peer-edge, and action-to-environment-return interventions; realized-exposure monitors; pre-intervention common-input measurements; cluster identities; repeated-cycle timestamps; a joint consequence endpoint; and leakage monitors. Existing sensor correlations cannot be relabeled to satisfy those requirements.



Frozen external role-intervention protocol

The protocol status is `FROZEN_PROTOCOL_NOT_EXECUTED`. No physical role outcome, assignment, or result has been invented. That unexecuted status is a substantive boundary and the exact external gate that remains after Contrast-and-Relation Study.

12.5 Reproducibility and finite formal boundary

Twenty-one deterministic application tests pass. They verify all parent-artifact hashes, contract and schema locks, the complete 2^{16} assignment enumeration, sharp-null control, mixed contrast decisions, exact noiseless relational recovery, many-to-one and partial representation, absent-role encoding, degradation under noise and missingness, split isolation, feature and parameter counts, trained model execution, adverse oracle comparison, preservation of Independent Replication Study and Joint-Role ADEMP Study counterresults, output determinism, overwrite refusal, unexecuted physical protocol status, and the final-test seal. Two complete runs reproduce ten scientific and figure artifacts byte for byte. The first execution's figure-directory omission is preserved as an implementation receipt; a post-outcome amendment created only the missing directory, and its five already-written scientific files match both complete runs exactly.

A bounded Lean leaf checks finite assignment-space, linear relational aggregation, absent-role, many-to-one, partial-weight, feature-count, and parameter-count invariants. Its status depends on successful compiler validation and the separate axiom audit; it does not prove randomization validity beyond the declared isolated design, noisy map recovery, trained-model performance, external role validity, biology, consciousness, or universal causation.

13. External-role feasibility and fixed-source audit

Contrast-and-Relation Study ended with a precise problem rather than a missing citation. A complete external test requires more than a system that contains sensors, peers, context, and action. It requires intervention records that distinguish those roles, realized-exposure measurements showing what was actually delivered, raw outcomes at the unit of analysis, and enough temporal or cluster structure to estimate uncertainty without pretending that correlated observations are independent. External Source Audit asks whether a fixed public source already provides that test.

The source search located strong partial analogues. Instead of merging their strengths into a fictional composite dataset, the audit asks what each source can and cannot identify on its own. Candidate discovery and the first DynaTeamTHOR archive inventory occurred before the final eligibility schema was written. The audit contract records that chronology explicitly. It is a deterministic source-classification exercise, not a prospectively registered outcome experiment. Its seven-source candidate set is bounded rather than logically exhaustive; a later qualifying archive would supersede the feasibility non-pass without changing any result reported for the sources actually audited.

13.1 A source passes only when the whole endpoint is observable

For source s and each required field $r \in \mathcal{R}$, let the coverage state be

$$c_{sr} \in \left\{0, \frac{1}{2}, 1\right\}, \qquad 0 = \text{absent}, \quad \frac{1}{2} = \text{partial}, \quad 1 = \text{complete}. \tag{63}$$

The twelve fields are fixed public identity, known license, raw unit outcomes, randomized bottom-up drive, randomized top-down boundary or routing, randomized lateral peer edge, randomized or replay-controlled action return, measured realized exposure, pre-input assignment information, clusters or timestamps, a joint endpoint, and leakage monitors. Full eligibility is conjunctive:

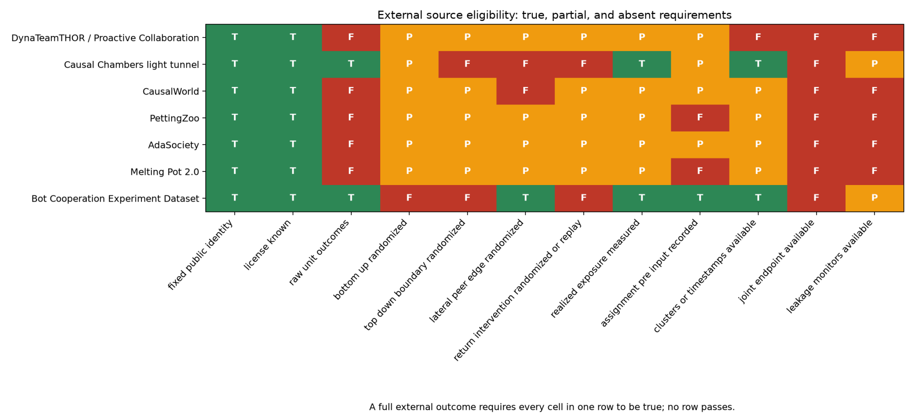
$$E_s = \prod_{r \in \mathcal{R}} \mathbf{1}(c_{sr} = 1). \tag{64}$$

A partial cell can make a source useful for architecture, feasibility, or one bounded causal role. It cannot be averaged with complete cells to pass the joint endpoint. This prevents a platform with many suggestive mechanisms but no raw outcome archive from outranking a narrower experiment with genuine randomized evidence, and it prevents separate papers from being combined as though their participants, assignments, and outcomes belonged to one study.

The causal data object needed for the joint analysis has the form

$$\mathcal{D}_s = \left\{Z_i^{BU}, Z_i^{TD}, Z_i^{LAT}, Z_i^{RET}, X_i^{pre}, E_i^{real}, C_i, t_i, Y_i, Q_i\right\}_{i=1}^{n_s}, \tag{65}$$

where the four Z variables are assigned roles, X^{pre} is pre-intervention common input, E^{real} records delivered exposure, C and t identify clusters and cycles, Y is the joint consequence, and Q contains fidelity and leakage monitors. When a source omits one or more of these components, a reduced estimand may remain possible, but the complete role contrast is not identified.



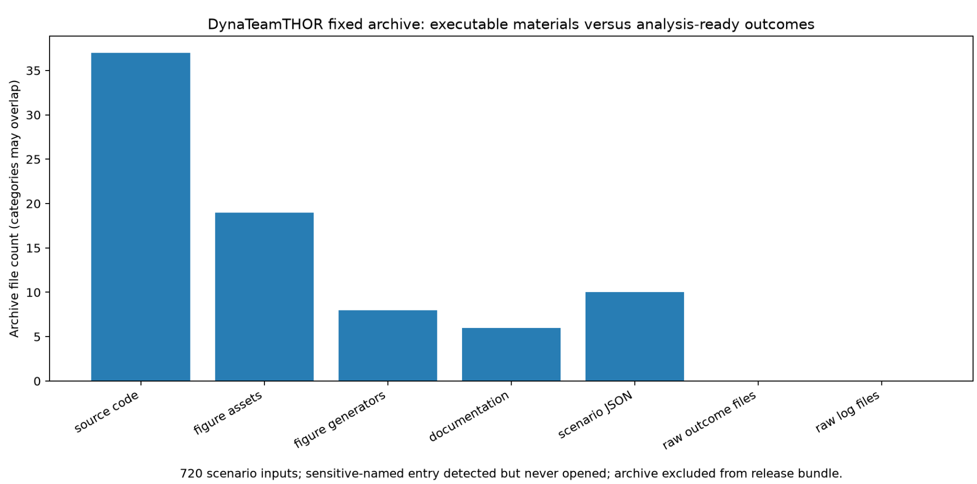
External source eligibility matrix

13.2 DynaTeamTHOR is the closest architecture match, not a completed external outcome

Li and colleagues' Proactive Collaboration study is the closest architecture located in this audit [59]. Its individual robots close a perception, action, and reflection loop; its communicative layer negotiates tasks and recruits or releases team members; and its experiments distinguish fixed, externally responsive, and team-controlled proactive collaboration. The published metrics include task success, partial success, temporal steps, and total action steps. This makes the system a substantive comparison for bottom-up sensing, top-down or boundary constraint, lateral communication, role reconfiguration, action, and environmental return.

That architectural overlap does not by itself satisfy Equation (65). The fixed Zenodo v1.0 archive is publicly identified by DOI 10.5281/zenodo.19240700, is licensed under MIT for the deposited code and data, contains 101 entries, and includes ten JSON scenario files with 72 scenarios each. The audit therefore verifies 720 scenario inputs. It also finds executable Python and Unity C-sharp source, figure-generating scripts, and embedded aggregate plotting values. It does not find raw per-episode outcome files, message or event logs, or a joint assignment table independently randomizing sensory drive, boundary, peer edge, and return or replay. The article's reported aggregate comparisons remain evidence for the authors' own questions; they are not silently converted into the different four-role estimand here.

One archive filename indicates an API-key environment file. The audit detects that name but never opens, extracts, hashes separately, or reports its content. The entire external ZIP is excluded from any paper release bundle. The paper records only its public custody, whole-archive hashes, structural inventory, and derived eligibility result.



DynaTeamTHOR archive inventory

13.3 Independent sources anchor different pieces without becoming one experiment

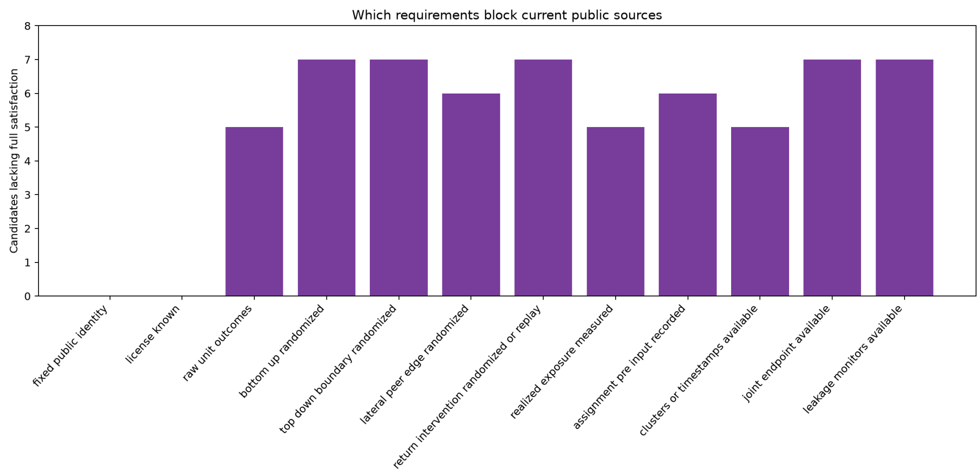
The audit includes six additional public candidates. Each strengthens a different part of the design while preserving the boundary between protocol capacity and observed outcome.

- The Causal Chambers light-tunnel archive used in Joint-Role ADEMP Study supplies raw physical intervention and time-series data and a valuable transfer stress test [56]. It does not implement top-down team boundaries, lateral peer-edge assignment, or action-return replay.
- CausalWorld provides a robotic manipulation environment designed for causal interventions and transfer [60]. It is a prospective implementation substrate, not a fixed raw four-role outcome archive, and it lacks a peer-edge

intervention.

- PettingZoo supplies a standard agent-environment-cycle interface for multi-agent environments [61]. An API can host the proposed experiment, but an API is not itself an assignment or outcome.
- AdaSociety combines adaptive physical surroundings with explicit and alterable social structures [62]. That makes it unusually relevant to the joined physical-and-social grammar, while its published benchmark does not expose the required full-factorial role archive.
- Melting Pot provides multi-agent social substrates and held-out social scenarios [63]. It supports prospective action-return and peer-interaction tests but does not supply the frozen joint intervention dataset.
- Shirado and Christakis randomized autonomous network interventions in public-goods groups and deposited raw data for 1,024 human participants in 64 networks [64]. This is the strongest independent lateral-role anchor in the audit: bots changed partner-selection topology and cooperation was measured. It does not jointly randomize sensory drive, top-down boundary, or action-return replay.

These are not failed papers. They answer different questions with different units, treatments, and outcomes. The non-pass applies only to the complete Contrast-and-Relation Study endpoint. The stronger scientific conclusion is therefore disaggregated: a fixed physical-intervention anchor exists; a raw lateral-network intervention anchor exists; several relevant multi-agent implementation platforms exist; and the exact crossed external endpoint remains unobserved.



Requirements blocking current public sources

13.4 What a qualifying external experiment would add

The closest feasible route is to implement the frozen protocol in a platform that exposes embodiment, communication, mutable team structure, and environmental consequences. One prospective DynaTeamTHOR extension would randomize: (1) a sensory-driver channel or controlled observation perturbation; (2) a boundary or routing rule governing task or team access; (3) a peer-message edge or matched communication exposure; and (4) live action-to-world return versus a preregistered matched replay or blocked-return condition. Scene or house should be the cluster; assignments must precede the perturbed inputs; actual messages, visibility, role changes, and action consequences must be logged; and success, partial success, time, action count, and a declared multivariate consequence should be retained at episode level.

That design must also measure spillover and leakage. A sensory perturbation that changes which peers are contacted is not an isolated bottom-up contrast unless the exposure path is represented. A team-boundary treatment that changes computational budget or model access is not capacity matched. A communication-edge assignment that is ignored or bypassed does not equal realized lateral exposure. A replay that changes latency, controllability, or information content is not a clean return-path control. These are precisely the distinctions that the controlled-study pipeline was built to preserve.

For the seven audited candidates,

$$\sum_{s=1}^{7} \mathbf{1}(E_s = 1) = 0. \tag{66}$$

None of the seven audited candidates passes the full conjunction. Equation (66) is a source-audit result, not evidence that the joined mechanism fails. It means the external endpoint has not yet been tested within this bounded candidate set by a source that meets the frozen admission rule. The admissible next step is to execute the physical protocol or obtain a fixed archive containing all elements of Equation (65), not to infer the missing intervention from correlations or aggregate plots.

13.5 Reproducibility and formal non-applicability

Twenty-one deterministic tests pass. They verify the contract, seven-source ledger, twelve required fields, whole-archive hashes and size, 101-entry inventory, ten scenario files, 720 scenario inputs, absence of analysis-ready raw outcome and log candidates, detection without inspection of the sensitive-named entry, release exclusion, zero full-role passes, explicit architecture-candidate status, preservation of the source-through-contrast artifacts, refusal to overwrite an existing output directory, and the sealed final test. Two authoritative executions reproduce eight tables, JSON results, and original figures byte for byte.

External Source Audit adds no Lean file. Equations (63)–(66) are definitions and a finite audit count already checked directly against the frozen ledger; formalizing their Boolean conjunction would not establish source truth, data availability, randomization, or transportability. The meaningful Contrast-and-Relation Study Lean invariants and compiler receipt remain unchanged. This explicit non-applicability decision prevents a decorative proof artifact from being mistaken for external empirical validation.

14. Independent cumulative and release-form audit

14.1 The cumulative result is mixed by design

The executed work consists of four synthetic studies, one bounded physical transfer check, and one fixed-source feasibility audit. Their results are not pooled into a single score. The first synthetic study verifies that the declared analysis can recover known support, practical equivalence, reversal, shared-input bias, leakage failure, mediation, abstraction stability, and exact-capacity parity. The second independently replicates the operation-order diagnostic and tests unseen channel remapping. The third crosses the three roles, evaluates the estimators through ADEMP, executes ordered and multivariate distances, stresses learned maps, compares stronger capacity alternatives, and checks a fixed physical archive without importing synthetic labels. The fourth isolates contrasts for exact randomization tests, replaces a forced permutation with a relational map, and measures the cost of learning that map against an oracle and trained alternatives.

The support and failure surfaces remain coequal. Four isolated contrasts pass the multiplicity-controlled rule; three remain unresolved; a null control is not rejected. A generic equivariant model ties the role-aligned model in one fixture; a larger flat interaction model catches up in another. Learned relational maps lose accuracy under missing probes and noise, and the oracle remains better. The fixed Causal Chambers measurements support out-of-distribution physical transfer only; they do not instantiate the crossed causal-role experiment. These outcomes jointly support the measurement architecture and several finite structural efficiencies while leaving SAN-specific predictive uniqueness, biological instantiation, and consciousness unestablished.

14.2 A four-source freshness pass

The first external audit examined seven fixed candidates. The independent freshness pass searched specifically for public collaboration datasets or protocols that add raw messages, actions, randomized facilitation or framing, peer interaction, and returned task consequences. Four sources warranted full strongest-form intake:

Additional source	Strongest directly supported contribution	Fields it adds	Why it does not close the crossed endpoint
How Communication Modalities Shape Topology in Generative Multi-Agent Systems [65]	A reproducible four-agent deliberation benchmark with 12 prompt-level conditions, 97 sessions, 3,869 raw messages, fixed seeds, prompt histories, and topology/content metrics.	top-down framing; peer messages; clustered sessions; structural endpoint	The public artifact does not provide independently randomized bottom-up drive, peer edge, and action-return conditions in one factorial, and the audited paper does not expose a stable named raw-data archive and license locator.
Alsobay et al. and GRAIL [66]	A preregistered study of 1,475 participants in 281 five-person groups, randomized among four facilitation conditions, with 14,343 messages and task outcomes; the MIT-licensed platform records server-side timestamps and complete interaction logs in new experiments.	randomized context/facilitation; peer messages; group clustering; decision endpoint; prospective logging platform	Facilitation is one randomized role. The published experiment does not independently randomize local input, peer edges, and return-path replay, and the platform is not itself the study's raw crossed outcome archive.
JUSThink [67]	A fixed CC BY 4.0 Zenodo corpus with event logs for 39 dyads, synchronized transcripts for ten dyads, pre/post responses, collaborative actions, and robot feedback.	actions; peer dialogue; timestamps; task return; dyad clusters	The corpus records a rich natural sequence but does not independently assign all four intervention roles or provide the required crossed exposure and leakage design.
Pow-Wow [68]	Temporally aligned observations, actions, and human communication in two-human versus two-agent Pommerman teams, plus a communication-policy comparison against noncommunicating agents.	local observations; peer communication; action; game consequence	Communication policy is not a crossed randomization of all four roles, and the audited public route did not supply a fixed licensed raw archive satisfying the admission contract.

Let $\mathcal{S}_8$ denote the original seven-source set and $\mathcal{F}_9$ the four-source freshness set:

$$\mathcal{S}_9 = \mathcal{S}_8 \cup \mathcal{F}_9, \qquad |\mathcal{S}_9| = 11. \tag{67}$$

Applying the unchanged twelve-field conjunction in Equation (65) gives

$$\sum_{s \in \mathcal{S}_9} \mathbf{1}(E_s = 1) = 0. \tag{68}$$

Equation (68) is a bounded availability result. It does not combine unrelated partial-role studies into a synthetic external experiment, and it does not infer a missing assignment or outcome. The new sources sharpen the implementation route: GRAIL can support controlled group-context studies; JUSThink and Pow-Wow show how peer communication, action, and returned task evidence can be logged; and the communication-topology benchmark shows that framing manipulations can reorganize artificial-agent interaction. A qualifying execution must still cross the roles prospectively under one source identity and one uncertainty model.

14.3 Nonduplication and contribution boundary

The fifteen-paper program contains neighboring work on cross-scale approximation, shared-context agents, cellular-to-population reinstatement, and Shannon/PWD information. This paper remains distinct because its irreducible deliverable is not another general account of hierarchy or recurrence. It supplies a role-indexed intervention grammar, micro–macro commutation tests, explicit macro-intervention fibers, realization-mixture and operation-order stress tests, interference-aware exposure maps, longitudinal role migration, and the same discriminating contract across neural, collective-animal, and artificial systems. Neighboring mechanisms enter only when they instantiate or challenge one of those causal roles.

14.4 Losslessness, source fidelity, and proof boundary

The cumulative claim audit preserves all 135 claims registered across the six prior claim matrices and adds the four-source freshness delta rather than replacing earlier rows. Historical SAN claims remain tied to their dated source class; established results remain tied to external primary or authoritative sources; synthetic results remain tied to frozen contracts and exact outputs; and prospective hypotheses remain labeled as tests. The full-context and revision-target checks find no reader-facing correction that requires deleting a supported earlier claim. The release-form change is organizational: the abstract now states the scientific problem, mechanism, mixed evidence, and open endpoint without using revision history as a substitute for the argument. The detailed development record remains in Sections 9–14 and the companion ledgers.

No new Lean file is warranted at this gate. The meaningful finite invariants remain the twelve relational and factorial theorems already compiler checked. Equations (67)–(68) summarize a declared source ledger; formalizing that finite count would not verify source identity, licensing, randomization, exposure, or external validity. The sealed final test remains unopened.

15. Falsification and failure conditions

The framework is designed to fail visibly under the following conditions.

1. **Micro sufficiency:** one low-level model predicts every claimed macro effect and every intervention without loss relevant to the task.
2. **Macro readout only:** changing the supposed macro variable has no implementable intervention distinct from relabeling local state.
3. **Lateral confounding:** peer effects disappear after shared input, environment, and selection are controlled.
4. **Intervention leakage:** the intended role cannot be manipulated without uncontrolled changes larger than the estimated target effect.
5. **Coarse-graining fragility:** small reasonable changes in $\boldsymbol{\tau}$ reverse the result.
6. **Noncommutation:** the high-level model fails to reproduce mapped low-level intervention effects beyond the preregistered tolerance.
7. **No predictive residual:** PWD-aware or three-role models do not outperform established rate, phase, message-passing, causal-abstraction, or recurrent baselines.
8. **No return dependence:** removing environmental or bodily consequence does not alter the claimed recurrent agency outcome.
9. **Time instability without detection:** causal roles migrate during learning but the model treats them as fixed.
10. **Claim-class inflation:** collective order, task performance, or application success is used as evidence of consciousness without an independent consciousness test.
11. **Intervention-fiber instability:** low-level realizations assigned to one macro intervention produce materially different macro outcomes, but the model reports a single effect without the realization distribution.
12. **Unmodeled operation-order dependence:** marginalization and abstraction yield meaningfully different results, but the selected order is not declared or stress-tested.
13. **Unsupported exposure:** the treatment or peer-exposure contrast lacks positivity in the target population.
14. **Exposure-map fragility:** a reasonable alternative interference mapping reverses the role effect and the experiment cannot discriminate between mappings.
15. **Assignment–realization collapse:** imperfect or heterogeneous realization is hidden by treating randomized assignment as the delivered mechanism.
16. **Post-treatment adjustment:** activity, load, or another descendant of the intervention is conditioned on without a valid mediation or longitudinal design.
17. **Tolerance inflation:** a practical bound is widened after calibration or held-out results reveal excessive noise.

18. **Underpowered null promotion:** statistical nonsignificance is labeled equivalence without an interval lying inside the preregistered equivalence region.
19. **Capacity unfairness:** SAN receives more parameters, tuning, observations, compute, or action opportunities than the alternative used to claim an advantage.
20. **Selective outcome retention:** a null, adverse, reversed, minority, or capacity-sensitive result is omitted from the reported result ledger.
21. **External-evidence inflation:** a protocol platform, architectural analogy, aggregate plot, or partial-role dataset is reported as though it supplied the missing raw joint intervention outcome.

A negative result is recorded at the tested atom, preparation, intervention, scale, exposure mapping, outcome, and decision margin. It changes the parameterization or scope of that tested formulation while preserving the source genealogy and the remaining untested architecture. An unresolved result remains unresolved; it is neither evidence for equivalence nor evidence for absence.

16. Accountability in multiscale artificial systems

The same causal decomposition has practical value for governance. When an artificial system produces harm, “the model did it” and “one agent did it” may both be incomplete. A responsibility analysis should identify:

- developer-imposed objectives and constraints;
- institution-level deployment rules;
- shared-context or memory state;
- local agent actions;
- peer messages and coordination;
- tool permissions and execution pathways;
- environmental feedback;
- which actors had knowledge, control, and a feasible intervention.

Counterfactual replay can estimate which intervention would have changed the outcome. Provenance logs can show which state and source version were available. The analysis should report distributed contribution without dissolving responsibility into a vague system-level cause.

17. Discussion

17.1 What the framework adds

The general ideas of multilevel explanation, downward causation, causal abstraction, thalamocortical control, synchronization, and emergence all have substantial prior literatures. SAN’s contribution here is a joined operational architecture with a specific source genealogy:

1. receiver-conditioned local difference;
2. bottom-up construction of larger patterns;
3. lateral phase, message, and inhibitory coordination;
4. physically implemented contextual constraint;
5. action and returned consequence;
6. longitudinal change in the roles themselves.

The paper also supplies one common intervention grammar across neural, collective, and artificial systems. Similar mathematics does not imply identical implementation. The translation is useful because it forces each domain to identify units, peer links, context implementation, macro variables, actions, return paths, and failure conditions.

17.2 Why the macro level can matter without becoming mysterious

A task rule, group boundary, or shared context matters when it compresses a family of lower-level states into a variable that supports stable prediction and feasible intervention. Its causal force is realized through the lower-level implementation. A macro model can be more useful than a micro model because it reduces irrelevant detail, improves signal-to-noise ratio, exposes an invariant, or matches the intervention available to the experimenter. Equations (15)–(18) and (23)–(26) test that usefulness.

This avoids two symmetrical errors. One error treats the macro level as an extra force. The other assumes that because a macrostate supervenes on local state, it cannot add explanatory or control value. A causal abstraction earns its place by preserving interventions with acceptable error. It also earns its intervention label by disclosing which lower-level realizations instantiate the label, how those realizations are sampled, and whether marginalization changes the result before versus after abstraction.

This framework does not require every domain to choose one causal-emergence criterion. Effective information asks whether a coarse state can be more causally informative under a declared intervention distribution [6,37]. Endogenous coarse-graining asks how a system physically builds and uses aggregate regularities [39]. Information decomposition asks which macro information is decoupled, downward, or synergistic relative to component information [40]. Biological relativity asks investigators not to privilege a scale in advance [41]. SAN's three-role vector can be evaluated with each of these lenses while retaining its stricter requirement that the role be attached to a realizable intervention and returned consequence.

17.3 Why peer relations deserve their own role

Hierarchical language often hides same-level interaction. Yet inhibition among local populations, coordination among oscillators, messages among artificial agents, and alignment among animals can change outcomes without first being summarized as a macro command. Lateral coordination is therefore not a leftover category. It has its own topology, delay, direction, sign, mediators, and confounds.

The distinction also clarifies excessive synchronization. Perfect uniformity can reduce differentiated information and flexibility. Chimera states, splay relations, selective communication subspaces, and controlled inhibition show that useful organization may combine coordination with maintained difference. The relevant variable is not “more synchrony” but the task- and receiver-specific organization that changes consequences.

17.4 What the cumulative evidence resolves and what remains open

Controlled Reference Study preserves Prospective Analysis Plan's source recovery, causal architecture, identification assumptions, power plan, and immutable manuscript while adding the first executed application. Its mixed result remains intact. The frozen pipeline recovered known positive, null, and reversed role effects; blocked a favorable estimate when leakage failed; separated peer influence from common input; avoided ordinary post-treatment adjustment; recovered separately randomized interventional pathways; distinguished stable from failed abstractions; and retained strong capacity-matched rivals. One of seven primary endpoints remained unresolved, and equal-information predictor encodings tied.

Independent Replication Study asked two new questions under a contract frozen before any output. First, would the operation-order result replicate with an independently implemented generator, a new seed namespace, 16,384 observations per distribution, and the same 0.05 bound? The new stable fixture passed practical equivalence at total variation 0.000793 [0, 0.013753], while the order-dependent control failed at 0.34448 [0.33208, 0.35688]. This resolves the measurement question for the new fixture; it neither relabels Controlled Reference Study's unresolved row nor establishes commutation in an external system.

Second, would an explicit structural map improve transfer to unseen physical-channel assignments against strong alternatives? At 21 fitted parameters, the canonical representation beat the matched flat representation by 3.18246 MSE [3.05369, 3.31123]. Yet the generic equivariant representation tied it, and a 37-parameter flat interaction model recovered practical equivalence. The supported claim is therefore structural parameter efficiency for this finite remapping problem. The result does not support predictive uniqueness for SAN terminology, and it does not show that an unstructured architecture cannot learn the same function when supplied the required interaction capacity.

Joint-Role ADEMP Study closes those development gaps without deleting their failure surfaces. The blocked joint-role factorial identifies all seven declared coefficients under controlled truth, while ADEMP shows where the estimator is calibrated, where naive unit precision undercovers, and where small interactions remain weakly powered. The ordered and multivariate branches distinguish stable controls from changed controls. The map is now learned from noisy probes, and its degradation under missing information, corruption, and a non-bijective system is measured rather than hidden. The generic equivariant tie remains, while the Joint-Role ADEMP Study flat-cross and random-feature alternatives remain worse under the declared richer target. A fixed physical archive supplies an out-of-distribution sanity check but not a physical instantiation of the full role taxonomy. Twelve finite invariants are compiler checked; statistical and scientific propositions remain empirical.

Contrast-and-Relation Study then separates three challenges that Joint-Role ADEMP Study could only name. Contrast-specific paired experiments give exhaustive sharp-null tests without pretending that the joint factorial's non-sharp coefficient nulls are exact. Their four supportive, three unresolved, and one null-control outcomes remain together. A soft relational map represents many-to-one, partial, and absent roles, but missing probes and noise reveal a steep recovery failure surface. The oracle's advantage over the learned relation directly measures the price of map error. The learned relation still beats the tested hard assignment, flat quadratic ridge, and two trained ReLU alternatives, but that bounded result coexists with Independent Replication Study's larger-flat recovery and Joint-Role ADEMP Study's equivariant tie.

External Source Audit converts the external-evidence question into a reproducible source map. DynaTeamTHOR demonstrates that all four architectural roles can coexist in a serious embodied multi-robot program, yet the deposited materials do not expose the raw crossed intervention data required here. Causal Chambers and the Shirado–Christakis network experiment independently anchor physical intervention and lateral network manipulation, respectively, without being combined into one artificial experiment. CausalWorld, PettingZoo, AdaSociety, and Melting Pot clarify where the frozen design could be implemented. The zero-pass verdict is therefore informative about data and design availability, not a negative result for the joined mechanism.

The remaining gate is now concrete rather than rhetorical. The frozen external protocol needs a physical system with independently randomized driver, boundary/routing, peer-edge, and return-path interventions plus realized exposure, raw episode outcomes, cluster or timestamp structure, and leakage measurements. It has not been executed. The final 8,192-

block test stays sealed until the remaining development sequence fixes how this external evidence and the cumulative mixed results govern release. Every later release must preserve all cumulative support, null, reversal, unresolved, oracle-loss, generic-equivalence, map-failure, capacity-sensitive, and external-source non-pass outcomes.

18. Conclusion

Multiscale systems do not require vague causation. Local events can be tested for how they construct larger states. Context can be tested for how its physical implementation constrains lower transitions. Peers can be tested for how their coupling changes outcomes beyond shared input. Action can be tested for how returned consequences reorganize the next cycle.

Self-Aware Networks joins these roles into a recurrent causal architecture. Its empirical burden is explicit: the joined model must support realizable interventions, identified assignment and exposure effects, approximately consistent micro–macro mappings, distinct role effects, disclosed low-level realization mixtures, alternative-scale and operation-order robustness, longitudinal tracking, and better held-out prediction or control than capacity-matched alternatives.

Controlled Reference Study shows that the proposed bookkeeping can meet several parts of that burden under controlled truth. It also shows why the remaining conditions matter. A narrow interval can still miss a practical bound; a favorable endpoint can be invalidated by leakage; a predictive macro readout can lack intervention effect; recurrence can proceed without a causal return; and equal-information representations can tie exactly. These are not peripheral caveats. They are the discriminators that prevent the taxonomy from passing by construction.

Independent Replication Study adds a narrower positive result. In an independently implemented finite remapping environment, an explicit structural map transferred to unseen assignments with fewer fitted parameters than the matched raw-channel model. The generic equivariant tie shows that the transferable invariant is not owned by SAN vocabulary, while the larger flat model's recovery shows that the advantage is one of representation and resource allocation rather than unique expressivity. The independent operation-order replication also passed the unchanged practical bound without rewriting Controlled Reference Study's unresolved result.

Joint-Role ADEMP Study shows that the same program can estimate jointly assigned roles, evaluate its own interval procedure, handle ordered and multivariate outcomes, learn a finite role map, survive comparison with larger raw-feature families in a richer fixture, and cross into a fixed physical dataset without importing synthetic labels into that dataset. It also shows why those advances do not collapse into one victory claim. Small interactions were weakly powered, the one-realization familywise diagnostic did not detect them, the generic equivariant model tied, non-bijective mapping failed by construction, and the physical transfer check did not instantiate the full causal taxonomy.

Contrast-and-Relation Study supplies exact isolated randomization tests, a non-bijective relational map class, and trained nonlinear alternatives. It does not make every planted effect detectable: lateral and two higher-order contrasts remain unresolved after multiplicity control. It does not make role discovery easy: absent-role detection and canonicalization deteriorate under noise and missing probes. It does not erase structural uncertainty: the oracle relation remains substantially better than the learned relation. These results refine what must be measured instead of converting model flexibility into automatic support.

External Source Audit shows that external evidence must be role complete before it can close the next gate. The closest architecture source contains the relevant loop but not the raw crossed outcome archive. Other sources provide genuine physical or lateral interventions but only for bounded subsets. Reporting those pieces separately is stronger than calling their union a completed test. The audit preserves their real contribution, specifies the missing data object, and leaves the physical role result explicitly unobserved.

Meeting the later empirical burden would show whether this structural efficiency and three-role causal accounting survive independently labeled physical role interventions and the sealed final evaluation. Practical equivalence delimits where added structure is unnecessary. An adverse or reversed effect localizes a failed prediction. Insufficient support or precision remains unresolved. Every outcome stays attached to its measured role, mapping, intervention family, exposure model, preparation, endpoint, and scale while the source genealogy and untested remainder remain intact. That is how the program turns multiscale causal language into cumulative evidence rather than a winner-take-all label.

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